Morgan Stanley | RESEARCH

IDEA

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Greater China Semiconductors | Asia Pacific

Old Memory: Peak Pricing Power?

Reduce exposure. Niche memory's pricing advantage is no longer enough. Rising cloud spending uncertainty, weaker consumer demand, and supply-demand normalization into 2028 shift the balance from earnings upside to downside protection.

Key Takeaways

- This report only expresses our view on niche memory (DDR3/4, SLC/MLC, NOR). It does NOT pertain to the global mainstream DDR5 or TLC/QLC NAND market.
- Pricing flexibility cuts both ways. Limited fixed-price LTAs supported faster price hikes vs. mainstream memory – but offer less protection as demand weakens.
- Demand risks are broadening. Cloud spending visibility is deteriorating; our PC semis work points to a deeper consumer correction.
- Scarcity should fade. Updated SLC NAND, MLC NAND, NOR and DDR4 models point to supply-demand normalization, particularly into 2028.

Fundamental snapshot – see the following section for a closer look.

Why we changed our minds – March caution, May upside surprise .

What has changed :

- Pricing power cannot offset weaker demand indefinitely.
- After the US midterms, our US team now expects ~20% of 2028e US data center capacity additions to shift to 2029.
- This could create an air pocket across the tech supply chain in 2028, potentially reducing pricing power and leading to an inventory build.

Supply-demand update – looking beyond the shortage .

Stock implications – downside support, not a reason to stay bullish:

- We downgrade PSMC, Winbond, GigaDevice, and Macronix to EW, downgrade Phison to UW, and lower price targets for AP Memory and Nanya Tech.
- We use 1.0x P/B as our downside valuation anchor, rather than treating peak earnings as a floor. Our July work highlighted book value support for memory IDMs; this is a downside framework, not a guaranteed trading floor or a uniform valuation methodology for fabless companies.
- Slower price hikes, order cuts, and inventory digestion are downside catalysts.
- Renewed cloud-order strength or delayed supply additions are the principal upside risks.

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific
Industry View **Attractive**

Company	Rating Change	Old PT	New PT
Winbond	OW -> EW	NT$288	NT$120
GigaDevice	OW -> EW	Rmb750	Rmb259
Powerchip	OW -> EW	NT$111	NT$54
Macronix	OW -> EW	NT$220	NT$90
Nanya Tech	OW	NT$600	NT$550
AP Memory	OW (Top Pick removed)	NT$1,666	NT$1,465
Phison	EW -> UW	NT$2,588	NT$1,357

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AlphaSignals Earnings Reaction

Impact to our thesis	Financial results versus consensus	Direction of next 12-month consensus EPS
Phison Electronics Corp (8299.TWO)		
Weakens our thesis	↓ Modest shortfall	↓ Modest revision lower
AP Memory Technology Corp (6531.TW)		
Unchanged	↑ Modest upside	↑ Modest revision higher

Source: Company data, Morgan Stanley Research

A Closer Look at the Fundamentals

Why we changed our minds – March caution, May upside surprise

We downgraded Winbond and Nanya Tech in March (Old Memory: Turning Selective) as elevated DDR4 pricing encouraged supply responses and customer pushback, pointing to narrower shortages and moderating price hikes. In May, we reversed those calls as lower expected supply from Micron and Hynix, alongside stronger server and enterprise SSD demand, widened the projected supply-demand gap (Old Memory: Upside Surprise Ahead). The distinction matters: March was a selective DDR4 call; today, we see a broader challenge to old memory's pricing durability.

What has changed

Pricing power cannot offset weaker demand indefinitely

Our constructive thesis was that niche memory, with limited fixed-price LTA constraints, could reprice faster and outperform mainstream memory. We now see that flexibility becoming a two-way risk. Rising uncertainty over cloud spending weakens confidence in incremental enterprise demand, while much weaker consumer demand challenges customers' ability to absorb further hikes.

After the US midterms, our US team now expects ~20% of 2028e US data center capacity additions to shift to 2029

This include some projects in Texas and Ohio, as governors introduce new data center legislation (refer to Buy the Drop: AI Investment Opportunities Post the US Midterms by Stephen Byrd). Stephen has shifted ~5GW of US data center capacity additions from 2028 to 2029 (Exhibit 11) – equivalent to 20% of US and 10% of global 2028 capacity additions. This implies a 4% Y/Y decline in US data center GW additions in 2028 before a 38% Y/Y rise in 2029. We now expect global data center additions to increase 9% in 2028 and 20% Y/Y in 2029.

This could create an air pocket across the tech supply chain in 2028, potentially reducing pricing power and leading to an inventory build

Our latest PC semis work highlights procurement pull-ins in 1Q–3Q26, followed by a deeper correction in 4Q, reinforcing the risk that earlier orders overstated underlying consumption (PC Semis: More Divergent Outlooks).

Supply-demand update – looking beyond the shortage

Our revised models for SLC NAND, MLC NAND, NOR, and DDR4 point to progressively more balanced markets, particularly into 2028. We now expect 2028 supply-demand deficits of -3%, -5%, and -1% for SLC NAND, MLC NAND, NOR, with a +1% excess for DDR4. Near-term tightness need not disappear immediately for stocks to de-rate: a narrowing shortage can reduce incremental pricing power before earnings peak. We therefore place

less weight on extrapolating recent ASP gains and more on normalized demand and supply responses and sustainable profitability.

Key Charts and Exhibits

Exhibit 1: What's changed

Company	EPS estimate changes	Reason for EPS estimate changes	Rating/price target changes	Reason for price target changes
Winbond	2026, 2027, 2028 EPS estimates revised +8%, -9%, and -22%	Incorporating 3Q26 actual results; changing dynamics in pricing power	Downgraded to EW; price target down from NT$288 to NT$120	Lower target P/BV multiple, reflecting segment volatility; slightly lower BV estimate
GigaDevice	2026, 2027, 2028 EPS estimates reduced 25%, 36%, and 33%	Changing dynamics in pricing power	Downgraded to EW; price target down from Rmb750 to Rmb259	Earnings estimate revisions; intermediate growth rate down to 8.0% from 11.3% to reflect changing dynamics in pricing power; beta raised to 1.50 from 1.14 to reflect segment volatility
Powerchip	2026, 2027, 2028 EPS estimates revised +10%, -11%, and -11%	Incorporating 2Q26 and 3Q26 actual results; changing dynamics in pricing power	Downgraded to EW; price target down from NT$111 to NT$54	Lower target P/BV multiple, reflecting segment volatility; slightly lower BV estimate
Macronix	2026, 2027, 2028 EPS estimates reduced 14%, 20%, and 21%	Changing dynamics in pricing power	Downgraded to EW; price target down from NT $220 to NT$90	Target P/BV multiple reduced to reflect segment volatility; BV estimate reduced
Nanya Tech	2026 EPS estimate unchanged; 2027 and 2028 EPS estimates lowered 7% and 9%	Changing dynamics in pricing power	Remain OW; PT down from NT $600 to NT$550	Target P/BV multiple reduced to reflect segment volatility; slightly lower BV estimate

AP Memory	2026, 2027, 2028 EPS estimates revised +21%, +2%, and -12%	Incorporating 3Q26 preliminary results; supply chain uncertainties into 2028	Remain OW but removed as Top Pick; price target down from NT$1,666 to NT$1,465	Earnings estimate revisions
Phison	2026, 2027, 2028 EPS estimates revised	Revised GM assumptions	Downgrade to UW; price target down from NT $2,588 to NT $1,357	Earnings estimate revisions, lower terminal growth rate, correction of prior RI model error

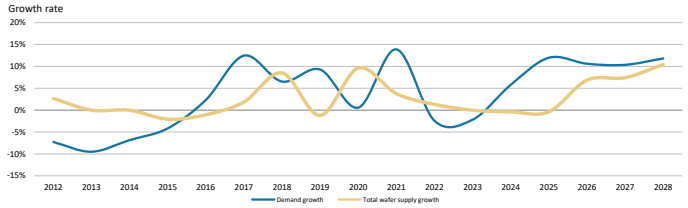
Source: Morgan Stanley Research

Exhibit 2: Order of preference: Greater China memory

	CXMT 688825.SS	AP Memory 6531.TW	Nanya Tech 2408.TW	Macronix 2337.TW	GigaDevice 603986.SS	PSMC 6770.TW	Winbond 2344.TW	Phison 8299.TWO
Rating	Overweight	Overweight	Overweight	Equal-weight	Equal-weight	Equal-weight	Equal-weight	Underweight
Trading Currency	CNY	TWD	TWD	TWD	CNY	TWD	TWD	TWD
Price Target	**88.0**	**1,465.0**	**550.0**	**90.0**	**259.0**	**54.0**	**120.0**	**1,357.0**
Current Price (as of 2026/11/10)	40.5	730.0	383.0	82.1	244.6	59.2	135.0	1,675.0
Upside/(Downside) (%)	117%	101%	44%	10%	6%	-9%	-11%	-19%
Market Cap (in USD mm)	411,514.4	3,605.6	36,029.8	4,522.5	24,464.2	7,567.2	17,223.7	10,427.1
Avg Daily Traded Vol (in USD mm)	2,947.9	106.1	1,006.6	395.4	2,115.4	395.9	722.6	366.5
Street View: Ratings								
Buy/Overweight	88%	100%	93%	80%	100%	100%	100%	73%
Hold/Equal-weight	6%	0%	7%	0%	0%	0%	0%	18%
Sell/Underweight	6%	0%	0%	20%	0%	0%	0%	9%
Bull Case Value	132.0	1,729.0	1,150.0	145.0	448.0	86.0	180.0	1,782.0
Upside (%)	226%	137%	200%	77%	83%	45%	33%	6%
Bear Case Value	40.0	500.0	300.0	41.0	112.0	27.0	50.0	655.0
Downside (%)	-1%	-32%	-22%	-50%	-54%	-54%	-63%	-61%
Risk/Reward Skew	183.0	4.3	9.2	1.5	1.5	0.8	0.5	0.1
Morgan Stanley Estimates								
FY26e	CNY	TWD	TWD	TWD	CNY	TWD	TWD	TWD
Sales	412,611	10,980	405,725	91,400	31,781	75,566	282,441	262,321
EBITDA	379,504	3,791	322,470	54,041	19,958	20,840	169,748	98,889
EBIT	340,684	3,730	310,209	47,525	19,708	13,276	154,137	98,819
EPS	3.28	21.39	75.96	20.17	25.87	6.23	26.96	419.09
FY27e								
Sales	718,168	20,918	688,388	169,350	49,301	102,808	434,309	209,990
EBITDA	644,196	6,878	543,822	122,315	25,444	38,093	275,243	56,117
EBIT	595,805	6,816	520,352	115,904	25,194	30,628	255,066	56,047
EPS	5.61	36.07	123.03	48.58	28.94	5.84	45.59	211.27
FY26 MSe vs. Consensus Mean								
Sales	10.8%	4.1%	2.4%	6.7%	14.6%	-1.6%	1.7%	-7.3%
EBITDA	13.1%	15.3%	4.1%	14.2%	25.7%	-53.5%	5.9%	-4.6%
EBIT	14.2%	18.2%	2.1%	13.2%	38.0%	-62.4%	7.4%	-3.4%
EPS	11.9%	12.1%	-0.6%	10.4%	28.5%	-10.8%	5.8%	-0.2%
FY27 MSe vs. Consensus Mean								
Sales	17.3%	15.9%	7.0%	19.2%	40.5%	-10.3%	-5.7%	-42.3%
EBITDA	15.3%	12.4%	6.0%	37.2%	46.4%	-32.1%	-13.9%	-28.2%
EBIT	20.1%	17.8%	1.7%	36.9%	57.2%	-37.6%	-6.6%	-37.3%
EPS	17.0%	12.3%	-3.0%	32.0%	30.7%	-32.6%	-12.6%	-38.1%
Valuation Multiples at Last Close								
FY26e								
P/E	12.3x	34.1x	5.0x	4.1x	9.5x	9.5x	5.0x	4.0x
EV/EBIT	17.5x	60.3x	4.5x	3.8x	7.5x	2.8x	3.4x	1.8x
EV/EBITDA	7.2x	27.9x	2.8x	3.1x	6.9x	5.2x	3.5x	2.5x
EV/Sales	6.6x	9.6x	2.2x	1.8x	4.3x	1.4x	2.1x	0.9x
FCF Yield	5.5%	2.0%	14.5%	-3.1%	10.4%	42.9%	3.5%	24.5%
FY27e								
P/E	7.2x	20.2x	3.1x	1.7x	8.5x	10.1x	3.0x	7.9x
EV/EBIT	4.0x	15.3x	1.0x	0.6x	5.0x	1.8x	1.5x	4.3x
EV/EBITDA	3.7x	15.2x	0.9x	0.5x	5.0x	1.4x	1.4x	4.3x
EV/Sales	3.3x	5.0x	0.7x	0.4x	2.6x	0.5x	0.9x	1.1x
FCF Yield	12.4%	3.1%	32.4%	70.7%	9.8%	24.0%	37.9%	15.5%
Implied Multiples on MS Price Target								
FY26e								
P/E	26.8x	68.5x	7.2x	4.5x	10.0x	8.7x	4.5x	3.2x
EV/EBIT	17.5x	60.3x	4.5x	3.8x	7.5x	2.8x	3.4x	1.8x
EV/EBITDA	15.7x	59.3x	4.4x	3.3x	7.4x	1.8x	3.1x	1.8x
EV/Sales	14.4x	20.5x	3.5x	2.0x	4.6x	0.5x	1.9x	0.7x
FY27e								
P/E	15.7x	40.6x	4.5x	1.9x	9.0x	9.2x	2.6x	6.4x
EV/EBIT	10.7x	35.3x	3.3x	1.4x	7.9x	7.6x	2.0x	5.1x
EV/EBITDA	9.9x	35.0x	3.2x	1.4x	7.8x	6.1x	1.9x	5.1x
EV/Sales	8.9x	11.5x	2.5x	1.0x	4.0x	2.3x	1.2x	1.4x

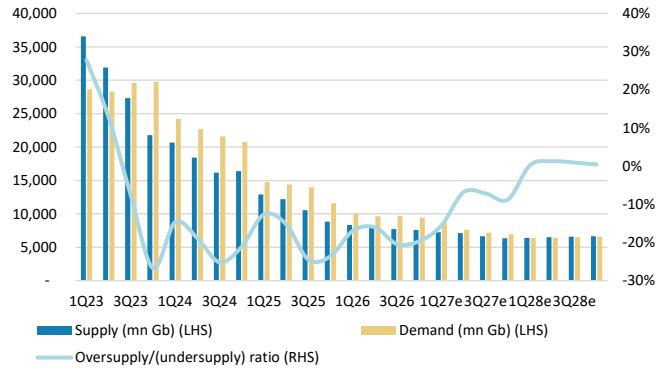
Source: FactSet (consensus mean), Morgan Stanley Research (e) estimates. Priced as of the close, November 10, 2026.

Exhibit 3: NOR flash demand and supply growth rates



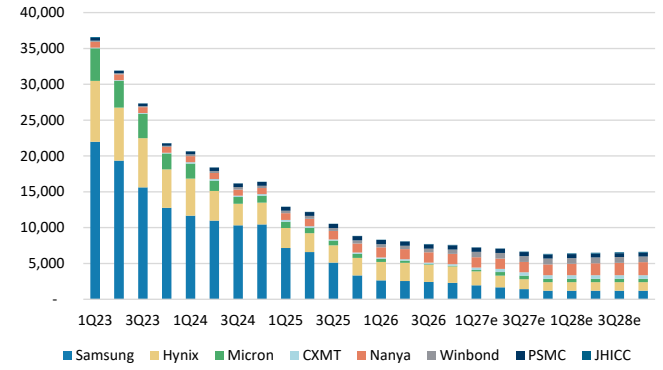
Source: Company data, Morgan Stanley Research

Exhibit 4: DDR4 quarterly supply and demand summary



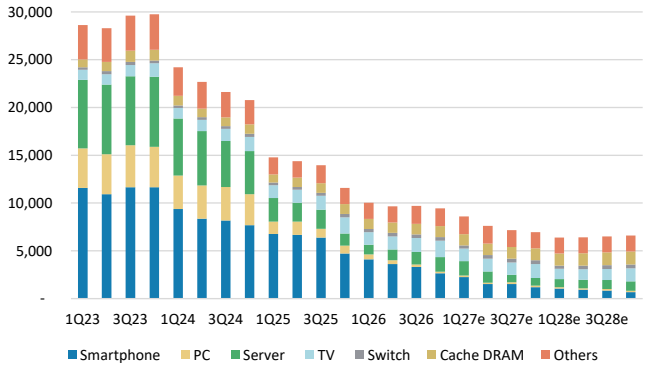
Source: Company data, IDC, Morgan Stanley Research (e) estimates

Exhibit 5: DDR4 quarterly supply breakdown (mn Gb)



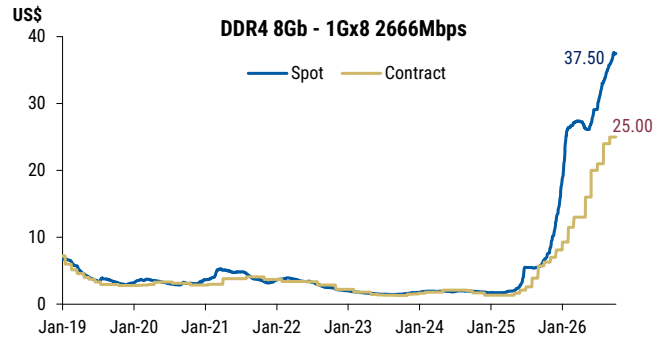
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 6: DDR4 quarterly demand breakdown by product (mn Gb)



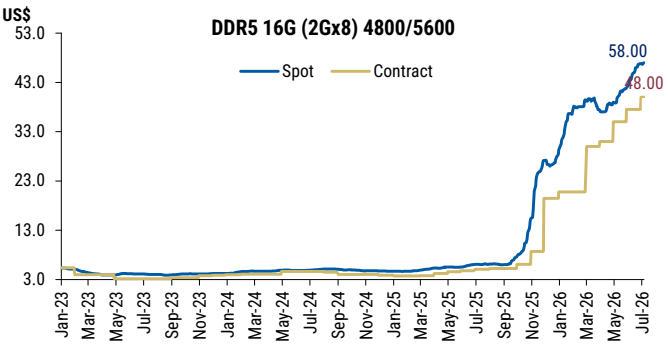
Source: IDC, Morgan Stanley Research (e) estimates

Exhibit 7: DDR4 8Gb (1Gx8) pricing chart



Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026.

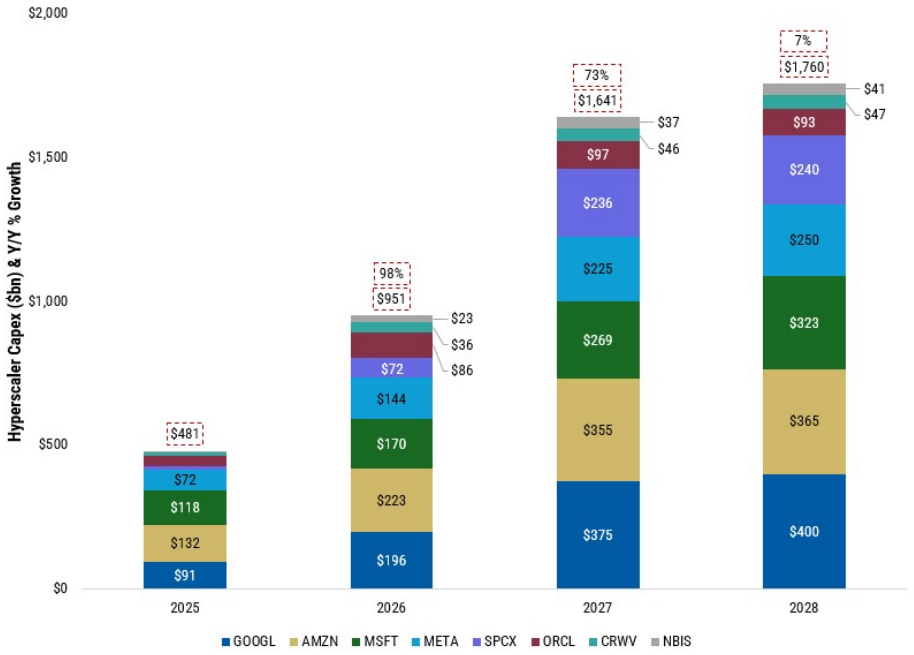
Exhibit 8: DDR5 16Gb (2Gx8) pricing chart



Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026.

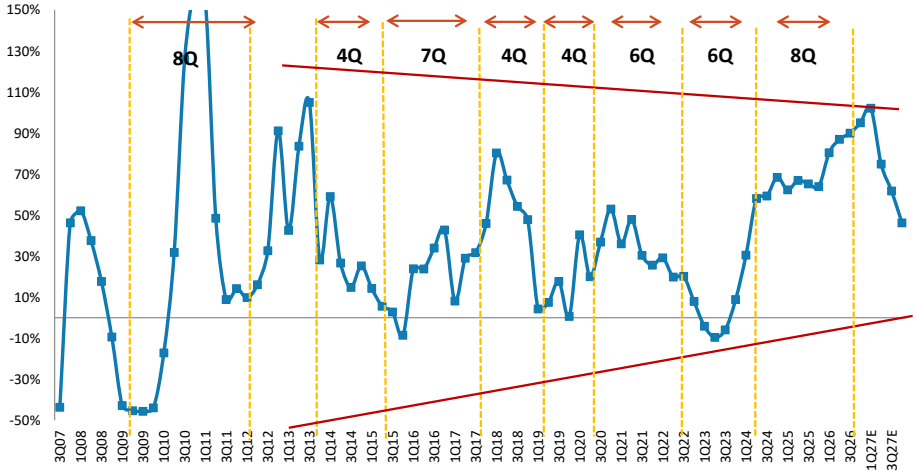
Exhibit 9:

Our hyperscaler capex forecasts remain unchanged at $1.6 trillion in 2027 and $1.8 trillion in 2028…with potential downward bias in 2028 if builds are delayed



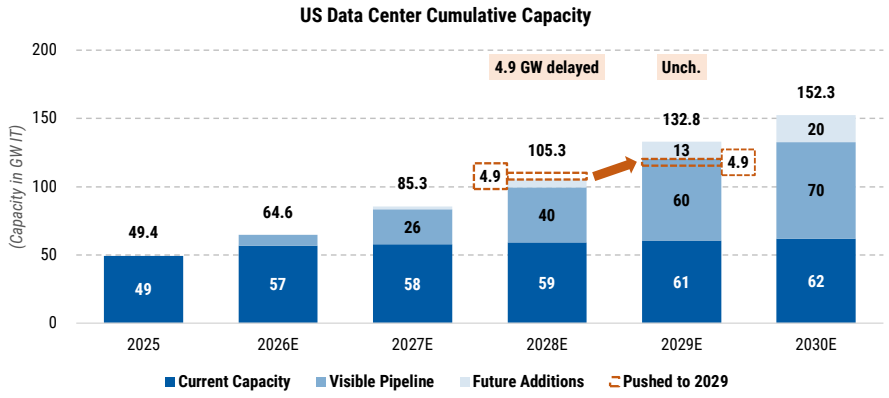
Source: Company data, Morgan Stanley US Research

Exhibit 10: US cloud capex was +90% Y/Y in 3Q26, up from +87% in 2Q26



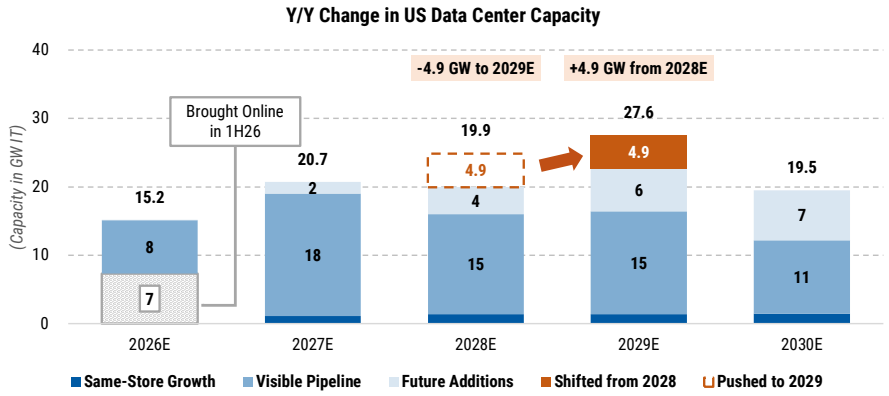
Source: Company data, Morgan Stanley US Research estimates; US cloud capex calculated from Alphabet, Microsoft, Amazon, and Meta

Exhibit 11: Our US team now expects~5 GW of capacity to be delayed from 2028 to 2029, reducing capacity estimates in 2028 while leaving 2029+ unchanged



Source: Morgan Stanley US Research estimates

Exhibit 12: Our US team now expects a y/y incremental 19.9 GW to come online in the US in 2028 and 27.6 GW in 2029; te build-out projection for 2030 is unchanged



Source: Morgan Stanley US Research estimates

Winbond: Estimate Revisions and Quarterly Financials

We adjust our EPS estimates +8% for 2026, -9% for 2027, and -22% for 2028: The 2026 EPS increase reflects 3Q26 actual results. We also factor in a potential peak in pricing power across NOR, SLC, and DDR4. Compared to the previous estimate, we now reduce our flash wafer ASP assumptions throughout 2027-2028 and 26% into 4Q28, and reduce our DRAM ASP per bit estimates throughout 2027-2028 and 23% into 4Q28, without changing our flash or DRAM shipment estimates. Apart from reduced revenue estimates, we also lower our gross margin estimates to account for the reduced ASP estimates. Our changes in net income/EPS estimates are largely a function of changes in our revenue and gross profit estimates. We believe it is a good time to make these changes now as a result of the narrower shortages, moderating price hikes, and potentially delays in 2028 US cloud capex.

Exhibit 13: Winbond: Estimate revisions

NT$ mn	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
Net sales	282,441	298,818	-5%	434,309	495,128	-12%	456,689	558,647	-18%
COGS	(87,364)	(119,579)		(133,945)	(169,180)		(149,883)	(180,294)	
Gross profit	195,076	179,239	9%	300,363	325,948	-8%	306,806	378,352	-19%
Operating expenses	(44,144)	(41,006)		(45,297)	(46,213)		(46,045)	(47,577)	
Operating profit	150,932	138,233	9%	255,066	279,735	-9%	260,761	330,775	-21%
Non-op. income (exp)	1,241	1,150		(33)	(33)		(53)	(53)	
Pretax Income	152,173	139,383	9%	255,033	279,701	-9%	260,707	330,722	-21%
Taxes	(30,234)	(27,726)		(51,007)	(55,940)		(52,141)	(67,331)	
Net income	122,622	112,009	9%	205,162	224,234	-9%	209,702	263,863	-21%
Reported EPS	**26.96**	**24.89**	**8%**	**45.59**	**49.83**	**-9%**	**46.60**	**59.96**	**-22%**
Margins									
Gross margin	69.1%	60.0%	9.1 ppt	69.2%	65.8%	3.3 ppt	67.2%	67.7%	-0.5 ppt
Operating margin	53.4%	46.3%	7.2 ppt	58.7%	56.5%	2.2 ppt	57.1%	59.2%	-2.1 ppt
Pretax margin	53.9%	46.6%	7.2 ppt	58.7%	56.5%	2.2 ppt	57.1%	59.2%	-2.1 ppt
Net margin	43.4%	37.5%	5.9 ppt	47.2%	45.3%	2.0 ppt	45.9%	47.2%	-1.3 ppt
Opex %	15.6%	13.7%	1.9 ppt	10.4%	9.3%	1.1 ppt	10.1%	8.5%	1.6 ppt

NT$	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
BVPS	**52.66**	**50.30**	**5%**	**98.30**	**100.18**	**-2%**	**144.95**	**160.19**	**-10%**

Source: Morgan Stanley Research (E) estimates

Exhibit 14: Winbond: Quarterly financials

NT$ in million	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2025	2026E	2027E	2028E
Total Revenues	**19,993**	**21,018**	**21,771**	**26,625**	**38,253**	**59,843**	**82,331**	**102,013**	**104,892**	**107,781**	**109,413**	**112,224**	**113,696**	**114,509**	**114,321**	**114,163**	**89,406**	**282,441**	**434,309**	**456,689**
Sequential Change	*7.0%*	*5.1%*	*3.6%*	*22.3%*	*43.7%*	*56.4%*	*37.6%*	*23.9%*	*2.8%*	*2.8%*	*1.5%*	*2.6%*	*1.3%*	*0.7%*	*-0.2%*	*-0.1%*				
Change vs Year Ago	*-0.6%*	*-2.2%*	*2.1%*	*42.4%*	*91.3%*	*184.7%*	*278.2%*	*283.1%*	*174.2%*	*80.1%*	*32.9%*	*10.0%*	*8.4%*	*6.2%*	*4.5%*	*1.7%*	*9.6%*	*215.9%*	*53.8%*	*5.2%*
Cost of Sales	(14,871)	(16,255)	(11,605)	(15,479)	(17,838)	(20,198)	(24,230)	(25,098)	(30,473)	(33,388)	(34,433)	(35,652)	(37,044)	(37,439)	(37,606)	(37,794)	(58,210)	(87,364)	(133,945)	(149,883)
Percent of Revenues	*74.4%*	*77.3%*	*53.3%*	*58.1%*	*46.6%*	*33.8%*	*29.4%*	*24.6%*	*29.1%*	*31.0%*	*31.5%*	*31.8%*	*32.6%*	*32.7%*	*32.9%*	*33.1%*	*65.1%*	*30.9%*	*30.8%*	*32.8%*
Gross Margin	**5,122**	**4,763**	**10,166**	**11,146**	**20,415**	**39,645**	**58,101**	**76,915**	**74,419**	**74,393**	**74,979**	**76,572**	**76,653**	**77,070**	**76,715**	**76,369**	**31,196**	**195,076**	**300,363**	**306,806**
Percent of Revenues	*25.6%*	*22.7%*	*46.7%*	*41.9%*	*53.4%*	*66.2%*	*70.6%*	*75.4%*	*70.9%*	*69.0%*	*68.5%*	*68.2%*	*67.4%*	*67.3%*	*67.1%*	*66.9%*	*34.9%*	*69.1%*	*69.2%*	*67.2%*
Total Opex	(6,087)	(6,058)	(6,463)	(7,054)	(7,865)	(10,666)	(14,317)	(11,296)	(11,079)	(11,208)	(11,432)	(11,578)	(11,562)	(11,536)	(11,494)	(11,453)	(25,662)	(44,144)	(45,297)	(46,045)
Percent of Revenues	*30.4%*	*28.8%*	*29.7%*	*26.5%*	*20.6%*	*17.8%*	*17.4%*	*11.1%*	*10.6%*	*10.4%*	*10.4%*	*10.3%*	*10.2%*	*10.1%*	*10.1%*	*10.0%*	*28.7%*	*15.6%*	*10.4%*	*10.1%*
Operating Income	(965)	(1,295)	3,703	4,092	12,550	28,979	43,784	65,619	63,340	63,185	63,547	64,994	65,090	65,534	65,220	64,917	5,534	150,932	255,066	260,761
Percent of Revenues	*-4.8%*	*-6.2%*	*17.0%*	*15.4%*	*32.8%*	*48.4%*	*53.2%*	*64.3%*	*60.4%*	*58.6%*	*58.1%*	*57.9%*	*57.2%*	*57.2%*	*57.1%*	*56.9%*	*6.2%*	*53.4%*	*58.7%*	*57.1%*
Total Non-operating Income(Loss)	(180)	(524)	(145)	(75)	138	1,039	77	(13)	(3)	(3)	(13)	(13)	(13)	(13)	(13)	(13)	(924)	1,241	(33)	(53)
Profit Before Taxes	(1,146)	(1,819)	3,558	4,017	12,688	30,018	43,861	65,605	63,336	63,182	63,534	64,980	65,077	65,520	65,207	64,903	4,610	152,173	255,033	260,707
Percent of Revenues	*-5.7%*	*-8.7%*	*16.3%*	*15.1%*	*33.2%*	*50.2%*	*53.3%*	*64.3%*	*60.4%*	*58.6%*	*58.1%*	*57.9%*	*57.2%*	*57.2%*	*57.0%*	*56.9%*	*5.2%*	*53.9%*	*58.7%*	*57.1%*
Taxes	159	188	(857)	(922)	(2,571)	(5,820)	(8,723)	(13,121)	(12,667)	(12,636)	(12,707)	(12,996)	(13,015)	(13,104)	(13,041)	(12,981)	(1,433)	(30,234)	(51,007)	(52,141)
Tax Rate	*13.9%*	*10.3%*	*24.1%*	*23.0%*	*20.3%*	*19.4%*	*19.9%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*31.1%*	*19.9%*	*20.0%*	*20.0%*
Reported Income (TW GAAP)	**(1,091)**	**(1,312)**	**2,943**	**3,422**	**10,114**	**24,317**	**35,422**	**52,768**	**50,953**	**50,830**	**51,111**	**52,268**	**52,346**	**52,700**	**52,449**	**52,207**	**3,962**	**122,622**	**205,162**	**209,702**
Percent of Revenues	*-5.5%*	*-6.2%*	*13.5%*	*12.9%*	*26.4%*	*40.6%*	*43.0%*	*51.7%*	*48.6%*	*47.2%*	*46.7%*	*46.6%*	*46.0%*	*46.0%*	*45.9%*	*45.7%*	*4.4%*	*43.4%*	*47.2%*	*45.9%*
Change vs Year Ago	*0.0%*	*0.0%*	*N.A.*	*0.0%*	*0.0%*	*0.0%*	*N.A.*	*N.A.*	*403.8%*	*109.0%*	*44.3%*	*N.A.*	*2.7%*	*3.7%*	*2.6%*	*N.A.*	*559.2%*	*2995.0%*	*67.3%*	*2.2%*
Reported EPS (NT$, TW GAAP)	**(0.24)**	**(0.29)**	**0.65**	**0.76**	**2.25**	**5.40**	**7.58**	**11.73**	**11.32**	**11.30**	**11.36**	**11.62**	**11.63**	**11.71**	**11.66**	**11.60**	**0.88**	**26.96**	**45.59**	**46.60**
Sequential Change	*61%*	*20%*	*-324%*	*16%*	*196%*	*140%*	*40%*	*55%*	*-3%*	*0%*	*1%*	*2%*	*0%*	*1%*	*0%*	*0%*	*493.9%*	*2961.9%*	*69.1%*	*2.2%*
Change vs Year Ago	*118%*	*N.A.*	*N.A.*	*N.A.*	*N.A.*	*N.A.*	*1059%*	*1442%*	*404%*	*109%*	*50%*	*N.A.*	*3%*	*4%*	*3%*	*N.A.*				
Book value per share	**21.8**	**21.0**	**22.5**	**25.4**	**27.6**	**33.0**	**46.0**	**52.7**	**64.0**	**75.3**	**86.7**	**98.3**	**109.9**	**0.0**	**13.8**	**13.4**	**25.4**	**52.7**	**98.3**	**145.0**

Source: Company data, Morgan Stanley Research (E) estimates

Winbond: Valuation Methodology

We cut our price target to NT$120

We continue to apply P/B multiple methodology to derive our price target, in line with our approach for Greater China memory IDM peers and in view of the industry's high earnings volatility. Our price target (base case scenario value) is set at 1.2x 2027e BVPS vs. our prior target multiple of 2.9x 2027e BVPS, to reflect the potential peak in pricing power. Our 1.2x target P/B is above our 1.0x P/B valuation anchor – we still favor the company's CUBE and S-SiCap businesses for their long-term potential. We see 1.0x P/B as the downside valuation anchor because book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns. When earnings are temporarily depressed, investors typically place greater weight on the underlying net asset base.

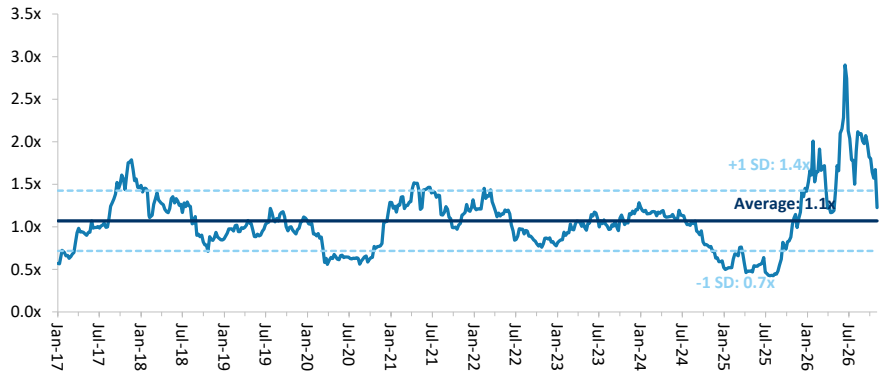
Our base case value/price target cut of 58% is almost entirely a function of reduced target multiple, rather than change in 2027e BVPS, which declines only 2% in our new estimate.

Our bull case value is reduced from NT$342 to NT$180, and our bear case value is reduced from NT$118 to NT$50, based on 1.8x and 0.5x 2027e BVPS, respectively (previously 3.4x and 1.2x).

We downgrade Winbond to EW

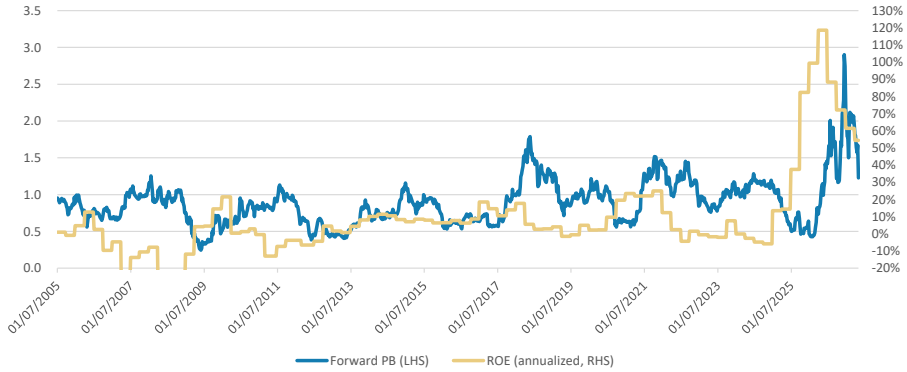
We continue to like the CUBE and S-SiCap businesses for longer-term potential, but we see near-term pressure from NOR, SLC NAND, and DDR4 businesses, in all of which industry supply is ramping faster than demand is increasing, especially as global cloud capex may face partial delay from 2028 to 2029. We believe the stock still has strong valuation support at 1.0-1.2x P/B, and therefore we believe an EW rating is fair.

Exhibit 15: Winbond: Historical forward P/B



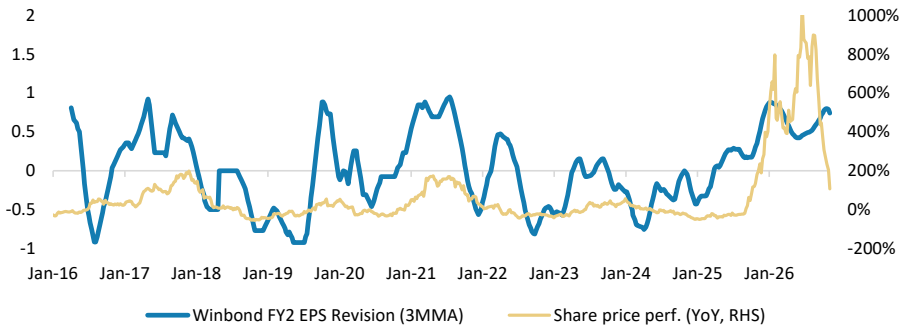
Source: Company data, Morgan Stanley Research estimates

Exhibit 16: Winbond: Historical Forward P/B vs. ROE



Source: Company data, Morgan Stanley Research estimates

Exhibit 17: Winbond: earnings estimate revision breadth



Source: Company data, Morgan Stanley Research estimates

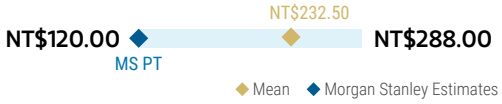
Risk Reward – Winbond Electronics Corp (2344.TW)

Pricing power may be peaking; still positive on CUBE and SiCap; EW

PRICE TARGET NT$120.00

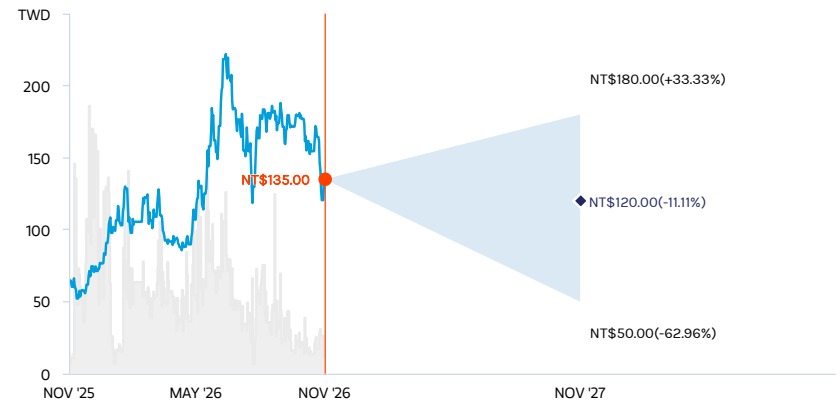
Base case, 1.2x 2027e P/B. This method is in line with the approach we adopt for Greater China memory IDM peers, in view of the industry's high volatility. Our target P/B is above our 1.0x P/B valuation anchor becaiuse we still favor the company's CUBE and S-SiCap businesses for their long-term potential.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



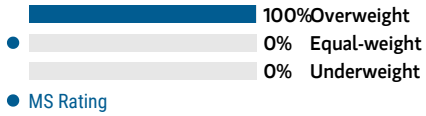
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Rising uncertainty over cloud spending weakens confidence in incremental enterprise demand, while much weaker consumer demand challenges customers' ability to absorb further price hikes
- Capacity expansion captures market upside but may create utilization or margin issues should pricing peak.
- We share management's view that CUBE could be meaningful in 2026, given engagement with multiple foundry partners and customers.
- We see potential for several headwinds for the logic business.
- Our price target implies 1.2x 2027e P/B, above our 1.0x P/B valuation anchor. Wwe still favor the company's CUBE and S-SiCap businesses for their long term potential

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Negative*

View descriptions of Risk Rewards Themes here

BULL CASE NT$180.00

1.8x 2027e BVPS

Stronger pricing trends: Flash wafer prices increase 30%+ Y/Y in 2027 and 10%+ Y/Y in 2028; DRAM ASP per bit rises 40%+ Y/Y in 2027 and 10%+ Y/Y in 2028.

BASE CASE NT$120.00

1.2x 2027e BVPS

Stabilizing ASP: Flash wafer prices increase 22% Y/Y in 2027 and decline 7% Y/Y in 2028; DRAM ASP per bit rises 25% Y/Y in 2027 and 1% Y/Y in 2028.

BEAR CASE NT$50.00

0.5x 2027e BVPS

Worse pricing trends than expected: Flash wafer prices increase 10% Y/Y in 2027 and decline 20%+ Y/Y in 2028; DRAM ASP per bit rises 10% Y/Y in 2027 and decline 10% Y/Y in 2028.

Risk Reward – Winbond Electronics Corp (2344.TW)

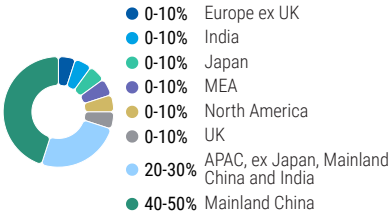
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
NOR ASP (%)	9	138	22	(7)
DRAM ASP (%)	(3)	273	25	1
Mobile RAM sales (NT$, mn)	5,302	22,533	36,116	37,440
DRAM sales (NT$, mn)	27,024	161,872	280,869	304,358
Flash sales (NT$, mn)	31,890	88,496	120,417	111,644

INVESTMENT DRIVERS

- DRAM pricing
- NOR pricing
- Technology migration

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- NOR Flash pricing increase, driven by stronger-than-expected demand
- Stronger-than-expected DRAM pricing given ongoing supply shortage
- Faster-than-expected SLC NAND development

RISKS TO DOWNSIDE

- NOR Flash down-cycle, driven by weaker demand
- Lower-than-expected DRAM pricing given ongoing oversupply
- Slower-than-expected SLC NAND development

OWNERSHIP POSITIONING

Inst. Owners, % Active 65.8%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	259,066	282,441 / 277,823	298,818
EBITDA (NT$, mn)	144,611	169,748 / 160,309	175,722
Net income (NT$, mn)	99,220	125,736 / 114,243	130,650
EPS (NT$)	22.05	26.96 / 25.47	29.03

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Winbond: Financial Summary

Exhibit 18: Financial summary

Income Statement

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	89,406	282,441	434,309	456,689
COGS	(58,210)	(87,364)	(133,945)	(149,883)
Depreciation Expense	(12,683)	(15,219)	(19,785)	(25,721)
Variable Cost	(26,224)	(52,437)	(94,368)	(99,840)
Gross profit	31,196	195,076	300,363	306,806
Operating expenses	(25,662)	(44,144)	(45,297)	(46,045)
Operating income	**5,534**	**150,932**	**255,066**	**260,761**
Non-operating income	(924)	1,241	(33)	(53)
Pre-tax income	4,610	152,173	255,033	260,707
Income tax	(649)	(29,552)	(49,871)	(51,005)
Reported net Income	**3,962**	**122,622**	**205,162**	**209,702**
Adj.wtd.avg.shrs(m)	4,500	4,500	4,500	4,500
Reported EPS (NT$)	**0.88**	**26.96**	**45.59**	**46.60**
Modelware EPS (NT$)	**0.88**	**26.96**	**45.59**	**46.60**

Balance Sheet

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	15,734	35,326	249,235	484,872
Mkt Securities	13,347	13,347	13,347	13,347
AR/NR	16,071	58,941	64,840	65,961
Inventory	25,758	109,835	120,828	122,916
Other	2,281	2,281	2,281	2,281
Current Assets	73,191	219,730	450,532	689,377
Long-term investments	19,266	19,073	19,013	18,973
Fixed assets	93,800	88,581	78,796	53,075
Other assets	5,935	5,935	5,935	5,935
Total Assets	192,192	333,319	554,275	767,360
S/T borrowings	2,739	2,739	2,739	2,739
AP/NP	8,224	17,313	24,593	26,071
Other ST liabilities	35,652	44,846	53,138	54,821
LT debt	26,399	26,399	26,399	26,399
Other LT liabilities	5,048	5,048	5,048	5,048
Total Liabilities	**78,062**	**96,345**	**111,918**	**115,078**
Common stocks	45,000	45,000	45,000	45,000
Preferred stocks	0	0	0	0
Capital Resaerve	13,752	13,752	13,752	13,752
Retained earning	49,184	172,028	377,412	587,336
Other shareholders' equity	0	0	0	0
Total Equity	114,130	236,974	442,358	652,282
Total Liab. & Shrhldr's Equity	**192,192**	**333,319**	**554,275**	**767,360**
BVPS	**25.4**	**52.7**	**98.3**	**145.0**

Cash Flow Statement

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	**11,183**	**38,349**	**232,666**	**244,393**
Net profits	3,177	122,622	205,162	209,702
Depreciation	12,308	15,219	19,785	25,721
Working Capital Change	(3,909)	(117,857)	(9,613)	(1,731)
Other adjustments	(394)	18,365	17,331	10,702
Cashflow from Investing	**(7,658)**	**(18,132)**	**(10,331)**	**9,649**
Capex	(5,296)	(10,000)	(10,000)	(10,000)
Change of LT Investment	0	0	0	0
Change of ST Investment	(971)	0	0	0
Other adjustments	(1,391)	(8,132)	(331)	19,649
Cashflow from financing	**(1,561)**	**(625)**	**(8,425)**	**(18,405)**
Increase in L/T debt	(1,698)	0	0	0
Increase in S/T debt	777	0	0	0
Cash Dividend Paid	0	0	0	0
Dir& Emp Bonus Paid	0	0	0	0
Other adjustments	(640)	(625)	(8,425)	(18,405)
Exchange rate adjustment	(330)	0	(0)	0
Net change in cash	**1,633**	**19,592**	**213,909**	**235,637**

Financial Ratio

	2025	2026E	2027E	2028E
Growth(%)				
Turnover	9.6	215.9	53.8	5.2
Margins (%)				
Gross Margin	34.9	69.1	69.2	67.2
Operating Margin	6.2	53.4	58.7	57.1
EBITA Margin	-8.0	48.1	54.2	51.5
Pretax Margin	5.2	53.9	58.7	57.1
Net Profit	4.4	43.4	47.2	45.9
Return (%)				
ROAE	4.3	113.6	88.9	48.1
ROAA	2.1	46.7	46.2	31.7
Gearing (%)				
Net Debt/Equity	31.9	7.1	(44.6)	(66.3)
Liabilities/Equity	68.4	40.7	25.3	17.6
Ratios (X)				
Current ratio	1.6	3.4	5.6	8.2
Quick ratio	0.7	1.5	3.9	6.6
Others				
AR/NR Turnover (days)	45	53	48	52
Inventory Turnover (days)	107	102	88	97
AP Turnover (days)	45	52	72	67
Cash Conversion (days)	107	104	64	82

Source: Company data, Morgan Stanley Research (E) estimates

GigaDevice: Estimate Revisions and Quarterly Financials

We cut our EPS estimates 25% for 2026, 36% for 2027, and 33% for 2028: The 2026e EPS change mainly reflects actual 2Q26 and 3Q26 results. The 2027e and 2028e EPS changes reflect a potential peak in pricing power across DDR4, NOR Flash, and SLC NAND. We now reduce our DRAM and others revenue estimates throughout 2027-2028 and 43% into 4Q28, mainly reflecting changes in our DDR4 ASP estimates. We also reduce our Flash (including NOR and SLC NAND) ASP estimates throughout 2027-2028 and 25% into 4Q28, without changing our Flash shipment estimates. We did not materially change our GM assumptions. Our changes in net income/EPS estimates are largely a function of changes in our revenue and gross profit estimates. We believe it is a good time to make these changes now as a result of the narrower shortages, moderating price hikes, and potential delay in 2028 US cloud capex.

Exhibit 19: GigaDevice: Estimate revisions

	Current	Previous		Current	Previous		Current	Previous	
Rmb mn	**2026E**	**2026E**	**Diff.**	**2027E**	**2027E**	**Diff.**	**2028E**	**2028E**	**Diff.**
Net sales	31,781	38,638	-18%	49,301	66,329	-26%	58,223	76,136	-24%
COGS	8,791	8,251		20,803	27,613		24,527	31,692	
Gross profit	22,990	30,387	-24%	28,499	38,715	-26%	33,696	44,443	-24%
Operating expenses	3,281	3,480		3,305	4,176		3,590	4,573	
Operating profit	19,708	26,907	-27%	25,194	34,539	-27%	30,106	39,870	-24%
Non-op. income (exp.)	757	(125)		(2,122)	20		(2,122)	20	
Pretax Income	20,465	26,782	-24%	23,072	34,559	-33%	27,984	39,890	-30%
Taxes	2,145	2,393		2,458	2,904		2,982	3,352	
Net income	18,236	24,345	-25%	20,504	31,610	-35%	24,892	36,493	-32%
Reported EPS (Rmb)	**25.87**	**34.72**	**-25%**	**28.94**	**45.09**	**-36%**	**35.13**	**52.05**	**-33%**
Margins									
Gross margin	72.3%	78.6%	-6.3%	57.8%	58.4%	-0.6%	57.9%	58.4%	-0.5%
Operating margin	62.0%	69.6%	-7.6%	51.1%	52.1%	-1.0%	51.7%	52.4%	-0.7%
Pretax margin	64.4%	69.3%	-4.9%	46.8%	52.1%	-5.3%	48.1%	52.4%	-4.3%
Net margin	57.4%	63.0%	-5.6%	41.6%	47.7%	-6.1%	42.8%	47.9%	-5.2%

Source: Morgan Stanley Research (E) estimates

Exhibit 20: GigaDevice: Quarterly financials

Rmb mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2025	2026E	2027E	2028E
Total Revenues	**1,909**	**2,241**	**2,681**	**2,372**	**4,188**	**7,378**	**8,732**	**11,483**	**11,643**	**12,024**	**12,516**	**13,119**	**13,489**	**14,126**	**14,882**	**15,727**	**9,203**	**31,781**	**49,301**	**58,223**
Sequential Change	*11.9%*	*17.4%*	*19.6%*	*-11.5%*	*76.6%*	*76.2%*	*18.4%*	*31.5%*	*1.4%*	*3.3%*	*4.1%*	*4.8%*	*2.8%*	*4.7%*	*5.4%*	*5.7%*				
Change vs Year Ago	*17.3%*	*13.1%*	*31.4%*	*39.0%*	*119.4%*	*229.2%*	*225.6%*	*384.2%*	*178.0%*	*63.0%*	*43.3%*	*14.2%*	*15.9%*	*17.5%*	*18.9%*	*19.9%*	*25.1%*	*245.3%*	*55.1%*	*18.1%*
Cost of Sales	1,194	1,412	1,589	1,307	1,798	2,467	2,591	1,936	4,913	5,071	5,288	5,530	5,669	5,955	6,283	6,621	5,502	8,791	20,803	24,527
Percent of Revenues	*63%*	*63%*	*59%*	*55%*	*43%*	*33%*	*30%*	*17%*	*42%*	*42%*	*42%*	*42%*	*42%*	*42%*	*42%*	*42%*	*60%*	*28%*	*42%*	*42%*
Gross Profit	**715**	**830**	**1,092**	**1,065**	**2,390**	**4,911**	**6,140**	**9,548**	**6,729**	**6,953**	**7,228**	**7,589**	**7,821**	**8,171**	**8,599**	**9,106**	**3,701**	**22,990**	**28,499**	**33,696**
Percent of Revenues	*37.4%*	*37.0%*	*40.7%*	*44.9%*	*57.1%*	*66.6%*	*70.3%*	*83.1%*	*57.8%*	*57.8%*	*57.7%*	*57.8%*	*58.0%*	*57.8%*	*57.8%*	*57.9%*	*40.2%*	*72.3%*	*57.8%*	*57.9%*
Incremental Margin	*73%*	*35%*	*60%*	*NM*	*73%*	*79%*	*91%*	*124%*	*-1769%*	*59%*	*56%*	*60%*	*63%*	*55%*	*57%*	*60%*	*49%*	*85%*	*31%*	*58%*
Total Opex	540	545	574	518	666	811	856	948	782	806	837	880	845	874	911	960	2,176	3,281	3,305	3,590
Percent of Revenues	*28.3%*	*24.3%*	*21.4%*	*21.8%*	*15.9%*	*11.0%*	*9.8%*	*8.3%*	*6.7%*	*6.7%*	*6.7%*	*6.7%*	*6.3%*	*6.2%*	*6.1%*	*6.1%*	*23.6%*	*10.3%*	*6.7%*	*6.2%*
R&D	292	276	292	257	358	406	441	461	290	300	315	335	290	300	315	335	1,117	1,666	1,240	1,240
Percent of Revenues	*15.3%*	*12.3%*	*10.9%*	*10.8%*	*8.6%*	*5.5%*	*5.0%*	*4.0%*	*2.5%*	*2.5%*	*2.5%*	*2.6%*	*2.1%*	*2.1%*	*2.1%*	*2.1%*	*12.1%*	*5.2%*	*2.5%*	*2.1%*
General & administrative	136	156	161	160	174	230	217	227	228	233	238	248	249	254	259	269	612	849	949	1,033
Percent of Revenues	*7.1%*	*6.9%*	*6.0%*	*6.7%*	*4.2%*	*3.1%*	*2.5%*	*2.0%*	*2.0%*	*1.9%*	*1.9%*	*1.9%*	*1.8%*	*1.8%*	*1.7%*	*1.7%*	*6.7%*	*2.7%*	*1.9%*	*1.8%*
Selling & marketing	112	113	121	101	134	175	198	260	263	272	283	297	305	320	337	356	446	766	1,116	1,318
Percent of Revenues	*5.8%*	*5.0%*	*4.5%*	*4.3%*	*3.2%*	*2.4%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*4.8%*	*2.4%*	*2.3%*	*2.3%*
Operating Income	**175**	**285**	**518**	**547**	**1,724**	**4,100**	**5,285**	**8,600**	**5,947**	**6,147**	**6,391**	**6,708**	**6,976**	**7,297**	**7,688**	**8,145**	**1,526**	**19,708**	**25,194**	**30,106**
Percent of Revenues	*9.2%*	*12.7%*	*19.3%*	*23.1%*	*41.2%*	*55.6%*	*60.5%*	*74.9%*	*51.1%*	*51.1%*	*51.1%*	*51.1%*	*51.7%*	*51.7%*	*51.7%*	*51.8%*	*16.6%*	*62.0%*	*51.1%*	*51.7%*
Change vs Year Ago	*12.7%*	*19.4%*	*36.9%*	*1331.9%*	*883.7%*	*1338.4%*	*920.2%*	*1471.3%*	*245.0%*	*49.9%*	*20.9%*	*-22.0%*	*17.3%*	*18.7%*	*20.3%*	*21.4%*	*88%*	*1192%*	*28%*	*19%*
Total Non-operating Income(Loss)	60	75	10	35	(103)	1,945	(543)	(543)	(530)	(530)	(530)	(530)	(530)	(530)	(530)	(530)	181	757	(2,122)	(2,122)
Profit Before Taxes	**236**	**360**	**528**	**583**	**1,621**	**6,045**	**4,742**	**8,057**	**5,417**	**5,617**	**5,861**	**6,178**	**6,445**	**6,766**	**7,157**	**7,615**	**1,706**	**20,465**	**23,072**	**27,984**
Percent of Revenues	*12%*	*16%*	*20%*	*25%*	*39%*	*82%*	*54%*	*70%*	*47%*	*47%*	*47%*	*47%*	*48%*	*48%*	*48%*	*48%*	*19%*	*64%*	*47%*	*48%*
Taxes	(4)	12	11	10	149	633	505	858	577	598	624	658	687	721	763	811	29	2,145	2,458	2,982
Tax Rate	*-1.7%*	*3.4%*	*2.1%*	*1.7%*	*9.2%*	*10.5%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*10.7%*	*1.7%*	*10.5%*	*10.7%*	*10.7%*
Net Income, Cont Ops	240	348	517	573	1,473	5,412	4,236	7,198	4,840	5,019	5,236	5,520	5,759	6,045	6,395	6,804	1,677	18,320	20,614	25,002
Percent of Revenues	*13%*	*16%*	*19%*	*24%*	*35%*	*73%*	*49%*	*63%*	*42%*	*42%*	*42%*	*42%*	*43%*	*43%*	*43%*	*43%*	*18%*	*58%*	*42%*	*43%*
Reported Income	**235**	**341**	**508**	**565**	**1,461**	**5,396**	**4,209**	**7,171**	**4,812**	**4,991**	**5,209**	**5,492**	**5,731**	**6,018**	**6,367**	**6,776**	**1,648**	**18,236**	**20,504**	**24,892**
Percent of Revenues	*12%*	*15%*	*19%*	*24%*	*35%*	*73%*	*48%*	*62%*	*41%*	*42%*	*42%*	*42%*	*42%*	*43%*	*43%*	*43%*	*18%*	*57%*	*42%*	*43%*
Change vs Year Ago	*15%*	*9%*	*61%*	*109%*	*523%*	*1483%*	*729%*	*1170%*	*229%*	*-7%*	*24%*	*-23%*	*19%*	*21%*	*22%*	*23%*	*49%*	*1007%*	*12%*	*21%*
Reported EPS (Rmb)	**0.35**	**0.51**	**0.76**	**0.85**	**2.08**	**7.70**	**5.94**	**10.12**	**6.79**	**7.04**	**7.35**	**7.75**	**8.09**	**8.49**	**8.99**	**9.56**	**2.48**	**25.87**	**28.94**	**35.13**
Change vs Year Ago	*15%*	*10%*	*61%*	*108%*	*490%*	*1399%*	*679%*	*1097%*	*226%*	*-8%*	*24%*	*-23%*	*19%*	*21%*	*22%*	*23%*	*50%*	*945%*	*12%*	*21%*

Source: Company data, Morgan Stanley Research (E) estimates

GigaDevice: Valuation Methodology

We cut our price target to Rmb259

We continue to derive our price target, which is our base case value, from a residual income model. We believe this best captures the stock's long-term value.

Our base case value cut of 65% can be attributed ~33% to earnings estimate revisions, as has been elaborated, and ~33% from multiple compression, driven by intermediate growth rate cut and beta increase. (Note: the overall price target change is calculated from the fully revised RI model. The individual contribution estimates are illustrative and should not be added together or compounded to derive the overall change.).

We cut our intermediate growth rate to 8.0% from 11.3% to reflect stronger changes in the supply environment compared to the end market demand over the medium to long term. The potential peak in pricing power for niche memory (DDR4, SLC, MLC, NOR) and the potential delay of 2028 US cloud capex now suggest slower revenue growth for GigaDevice from 2026 to 2036. We think a 3.3ppt p.a. growth rate cut reflects the new supply/demand dynamics with a closer gap.

We also raise our beta to 1.50 from 1.14, leading our cost of equity assumption to rise to 11.8% from 9.4%. This reflects a cautious valuation scenario, to account for the recent memory segment volatility, 70%+ correction from the stock price peak in June, and the fabless model, which does not provide a floor support for valuation similar to the IDM model. The elevated beta reflects the company's exposure to rapidly changing niche memory pricing dynamics that could severely affect earnings with a downward bias compared to our financial forecasts.

Our other key assumptions remain unchanged, including equity risk premium of 6.5%, risk-free rate of 2% and payout ratio of 40%.

Our bull case value falls to Rmb448 from Rmb1,300, and our bear case value to Rmb112 from Rmb324.

We downgrade GigaDevice to EW

We still see the company as a domestic leader in MCU localization and believe its customized DRAM business could show a revenue ramp over the next 1-2 years. We also still like its strategic partnership with CXMT, which provides both capacity support and process tech improvements. However, we see near-term pressure from the NOR, SLC NAND, and DDR4 businesses, in all of which industry supply is ramping faster than demand increase, especially as global cloud capex may face partial delay from 2028 to 2029. Fabless design houses do not face depreciation and utilization issues during a down-cycle, but at the same time they lack a similar valuation anchor as IDM/foundry players from their book value, and therefore we see potentially continued high volatility ahead.

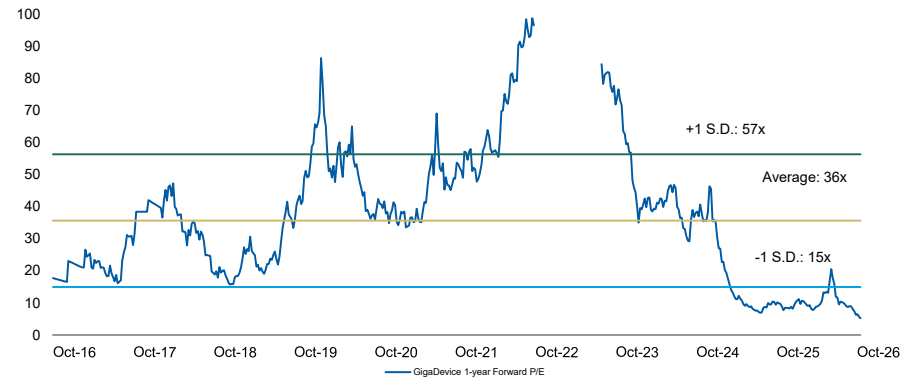
Our new price target implies a 2027e P/E of 9x, below the historical average of 38x since 2016, owing to changes in pricing power.

Exhibit 21: GigaDevice: Residual income model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	**36,965**	**51,998**	**70,739**	**87,895**	**106,425**	**126,437**	**148,050**	**171,391**	**196,601**	**223,826**	**253,230**	**284,987**
Net Profit	**18,236**	**20,504**	**24,892**	**26,883**	**29,034**	**31,357**	**33,865**	**36,574**	**39,500**	**42,660**	**46,073**	**49,759**
ROAE	**64.9%**	**46.1%**	**40.6%**	**33.9%**	**29.9%**	**26.9%**	**24.7%**	**22.9%**	**21.5%**	**20.3%**	**19.3%**	**18.5%**
Residual Income	10,219	12,696	14,981	15,664	15,938	16,157	16,342	16,506	16,656	16,797	16,934	17,069
Spread	**53.2%**	**34.3%**	**28.8%**	**22.1%**	**18.1%**	**15.2%**	**12.9%**	**11.1%**	**9.7%**	**8.5%**	**7.6%**	**6.7%**
Ending Equity Capital	36,965											
PV of Forecast Period	83,127											
PV of Continuing Value	62,274											
Equity Value	**182,366**											
No. of Shares	705											
Projected Price (Rmb)	**259**											

Source: Morgan Stanley Research (E) estimates

Exhibit 22: GigaDevice: Historical forward P/E



Source: Company data, Morgan Stanley Research estimates

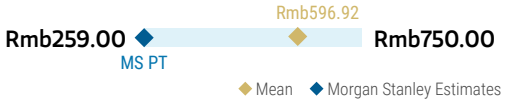
Risk Reward – GigaDevice Semiconductor Beijing Inc (603986.SS)

Cautious on peaking niche DRAM pricing power; EW

PRICE TARGET Rmb259.00

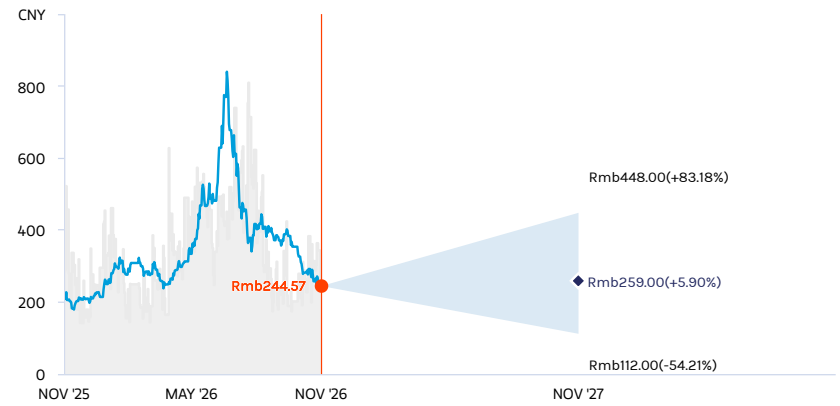
Base case value, derived from our residual income model. Key assumptions: cost of equity of 11.8% (beta 1.5, risk-free rate 2%, and risk premium 6.5%), a payout ratio of 40%, medium-term growth rate of 8.0%, and a terminal growth rate of 2.5%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



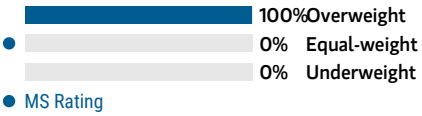
Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect GigaDevice to continue to gain share in China's MCU market. Auto MCUs offer bigger opportunities as local self-sufficiency is low.
- Rising uncertainty over cloud spending weakens confidence in incremental enterprise demand, while much weaker consumer demand challenges customers' ability to absorb further hikes
- We believe GigaDevice will ramp up 16G LPDDR4 and 32G DDR4 at CXMT's Gen 4 platform in 2027.
- Our price target implies 2027e P/E of 9x, below the historical average of 38x from 2016, reflecting multiple compression amid growth uncertainties.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power:	*Negative*
Secular Growth:	*Negative*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb448.00

15x 2027e EPS

NOR, SLC, and DRAM prices continue rising into 2027-28; MCU growth accelerates; meaningful WoW contribution: 1) Flash sales rise 80% and 20% Y/Y in 2027 abd 2028; 2) MCU rises 30% and 20% in 2027 and 2028; 3) meaningful contribution from WoW into 2028; 4) DRAM and others grows 60% and 40% in 2027 and 2028.

BASE CASE Rmb259.00

9x 2027e EPS

NOR, SLC, and DRAM prices stabilizes into 2027/28; MCU growth continues; moderate WoW contribution: 1) Flash sales rise 54% Y/Y in 2027 and 0% Y/Y in 2028; 2) MCU rises 16% and 23% in 2027 and 2028%; 3) moderate contribution from WoW into 2028; 4) DRAM and others rises 36% in 2027 but drops 7% in 2028.

BEAR CASE Rmb112.00

4x 2027e EPS

NOR, SLC, and DRAM prices decline in 2027-28; MCU growth stalls; WoW contribution delays: 1) Flash sales grow 20% Y/Y in 2027 but drop 10% Y/Y in 2028; 2) MCU rises 10% in 2027 but 0% in 2028; 3) limited contribution from WoW into 2028; 4) DRAM and others grows 20% in 2027 but drops 20% in 2028.

Risk Reward – GigaDevice Semiconductor Beijing Inc (603986.SS)

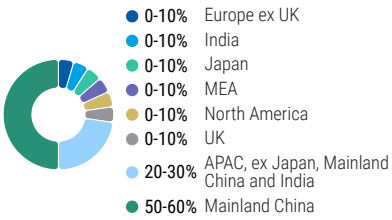
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
NOR Flash ASP growth (%)	4	159	36	0
MCU ASP growth (%)	(8)	0	0	0
Gross margin (%)	40	77	58	58

INVESTMENT DRIVERS

- Beneficiary of MCU localization
- Decreasing opex burden
- Development of AI glasses market

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

3/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- NOR up-cycle driven by stronger demand
- Superior chip design, leading to increasing exposure to low-density NOR
- Faster-than-expected DRAM development

RISKS TO DOWNSIDE

- NOR down-cycle driven by weaker demand
- Inferior chip design, leading to increasing exposure to mid-/high-density NOR
- Slower-than-expected DRAM development

OWNERSHIP POSITIONING

Inst. Owners, % Active 86.6%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (Rmb, mn)	18,970.0	27,731.3	31,780.8
EBITDA (Rmb, mn)	12,366.0	15,875.4	19,958
Net income (Rmb, mn)	6,586.0	13,702.5	18,236
EPS (Rmb)	8.9	20.1	25.9

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

GigaDevice: Financial Summary

Exhibit 23: Financial Summary

Income Statement

Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	9,203	31,781	49,301	58,223
COGS	(5,502)	(8,791)	(20,803)	(24,527)
Gross profit	3,701	22,990	28,499	33,696
Operating expenses	(2,176)	(3,281)	(3,305)	(3,590)
Operating income	**1,526**	**19,708**	**25,194**	**30,106**
Non-operating income	181	757	(2,122)	(2,122)
Pre-tax income	1,706	20,465	23,072	27,984
Income tax	29	2,145	2,458	2,982
Minority interests	(29)	(83)	(110)	(110)
Reported net Income	**1,648**	**18,236**	**20,504**	**24,892**
Adj.wtd.avg.shrs(m)	666	705	709	709
MW EPS (Rmb)	2.48	25.87	28.94	35.13
Reported EPS (Rmb)	**2.48**	**25.87**	**28.94**	**35.13**

Balance Sheet

Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	9,186	25,615	36,135	53,392
Mkt Securities	102	102	102	102
AR/NR	199	776	1,203	1,421
Inventory	3,066	4,180	9,892	11,663
Other	873	873	873	873
Current Assets	**13,425**	**31,545**	**48,205**	**67,450**
Long-term investments	4,085	4,085	4,085	4,085
Fixed assets	1,325	1,325	1,325	1,325
Deferred assets	432	432	432	432
Other assets	2,129	2,129	2,129	2,129
Total Assets	**21,397**	**39,516**	**56,176**	**75,421**
S/T borrowings	200	200	200	200
AP/NP	813	1,191	2,818	3,322
Other ST liabilities	934	934	934	934
LT debt	0	0	0	0
Other LT liabilities	226	226	226	226
Total Liabilities	**2,174**	**2,552**	**4,178**	**4,683**
Common shares	668	668	668	668
Additional capital	9,201	9,201	9,201	9,201
Retained earning	8,407	26,149	41,182	59,923
Other shareholders' equity	946	946	946	946
Total Equity	**19,223**	**36,965**	**51,998**	**70,739**
Total Liab. & Shrhldr's Equity	**21,397**	**39,516**	**56,176**	**75,421**

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Cash Flow Statement

Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	**2,032**	**17,173**	**16,242**	**23,657**
Net profits	1,648	18,236	20,504	24,892
Depreciation	250	250	250	250
Working Capital Change	(440)	(1,313)	(4,512)	(1,484)
Other adjustments	574	0	0	0
Cashflow from Investing	**(1,391)**	**(250)**	**(250)**	**(250)**
Capex	(1,049)	(250)	(250)	(250)
Change of LT Investment	(1,886)	0	0	0
Change of ST Investment	0	0	0	0
Other adjustments	1,544	0	0	0
Cashflow from financing	**(517)**	**(494)**	**(5,471)**	**(6,151)**
Increase in L/T debt	0	0	0	0
Increase in S/T debt	(698)	0	0	0
Cash Dividend Paid	(243)	(494)	(5,471)	(6,151)
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	516	0	0	0
Other adjustments	(92)	0	0	0
Exchange rate adjustment	21	0	0	0
Net change in cash	**146**	**16,429**	**10,521**	**17,256**

Financial Ratios

	2025	2026E	2027E	2028E
Growth(%)				
Turnover	25.1	245.3	55.1	18.1
Operating profits	88.1	1191.9	27.8	19.5
EPS	49.6	944.9	11.8	21.4
Margins (%)				
Gross Margin	40.2	72.3	57.8	57.9
Operating Margin	16.6	62.0	51.1	51.7
Pretax Margin	18.5	64.4	46.8	48.1
Net Margin	17.9	57.4	41.6	42.8
Return (%)				
ROAE	9.2	64.9	46.1	40.6
ROAA	8.1	59.9	42.9	37.8
Gearing (%)				
Net Debt/Equity	(46.7)	(68.8)	(69.1)	(75.2)
Liabilities/Equity	11.3	6.9	8.0	6.6
Ratios (X)				
Current ratio	6.9	13.6	12.2	15.1
Quick ratio	4.8	11.3	9.4	12.3
Others				
AR/NR Turnover (days)	9	9	9	9
Inventory Turnover (days)	174	174	174	174
AP Turnover (days)	49	49	49	49
Cash Conversion (days)	133	133	133	133

PSMC: Earnings Estimate Revisions

We adjust our EPS forecasts +10% for 2026 and -11% each for 2027 and 2028: The 2026e EPS change mainly reflects actual 2Q26 and 3Q26 results. The 2027e and 2028e EPS changes reflect a potential peak in pricing power across DDR3/3 and SLC NAND. We now reduce our DRAM wafer ASP estimates throughout 2027-2028 and 19% into 4Q28, without changing our DRAM wafer shipment estimates. We did not materially change our GM assumptions. Our changes in net income/EPS estimates are largely a function of changes in our revenue and gross profit estimates. We believe it is a good time to make these changes now as a result of the narrower shortages, moderating price hikes, and potential delay in 2028 US cloud capex.

Exhibit 24: PSMC: Earnings estimate revision summary

(NT$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	75,566	77,163	-2%	102,808	107,846	-5%	116,499	123,668	-6%
Gross profit	22,998	18,628	23%	40,866	43,266	-6%	49,738	53,158	-6%
Operating profit	30,584	26,017	18%	33,592	37,386	-10%	41,767	46,559	-10%
Pretax income	31,407	27,772	13%	34,408	38,071	-10%	42,583	47,244	-10%
Net income	28,060	25,164	12%	27,526	30,457	-10%	34,066	37,795	-10%
EPS (NT$)	**6.23**	**5.66**	**10%**	**5.84**	**6.54**	**-11%**	**7.23**	**8.11**	**-11%**
Margins									
Gross margin	30.4%	24.1%		39.8%	40.1%		42.7%	43.0%	
Operating margin	40.5%	33.7%		32.7%	34.7%		35.9%	37.6%	
Pretax margin	41.6%	36.0%		33.5%	35.3%		36.6%	38.2%	
Net margin	37.1%	32.6%		26.8%	28.2%		29.2%	30.6%	

(NT$)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
BVPS	43.2	43.1	0.3%	53.5	54.3	-1.5%	65.2	66.7	-2.3%

Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 25: PSMC: Quarterly financials

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Total Revenues	**11,116**	**11,278**	**11,840**	**12,495**	**13,572**	**17,290**	**21,342**	**23,362**	**24,292**	**25,447**	**26,195**	**26,875**	**27,818**	**28,820**	**29,587**	**30,275**	**46,730**	**75,566**	**102,808**	**116,499**
Sequential Change	*-0.1%*	*1.5%*	*5.0%*	*5.5%*	*8.6%*	*27.4%*	*23.4%*	*9.5%*	*4.0%*	*4.8%*	*2.9%*	*2.6%*	*3.5%*	*3.6%*	*2.7%*	*2.3%*				
Change vs Year Ago	*2.7%*	*1.4%*	*1.6%*	*12.2%*	*22.1%*	*53.3%*	*80.2%*	*87.0%*	*79.0%*	*47.2%*	*22.7%*	*15.0%*	*14.5%*	*13.3%*	*12.9%*	*12.7%*	*4.5%*	*61.7%*	*36.1%*	*13.3%*
Cost of Sales	(11,653)	(12,300)	(12,614)	(11,701)	(12,184)	(12,401)	(11,194)	(16,789)	(14,774)	(15,398)	(15,819)	(15,951)	(15,997)	(16,529)	(16,935)	(17,300)	(48,267)	(52,568)	(61,942)	(66,761)
Percent of Revenues	*104.8%*	*109.1%*	*106.5%*	*93.6%*	*89.8%*	*71.7%*	*52.5%*	*71.9%*	*60.8%*	*60.5%*	*60.4%*	*59.4%*	*57.5%*	*57.4%*	*57.2%*	*57.1%*	*103.3%*	*69.6%*	*60.2%*	*57.3%*
Gross Profit	**(536)**	**(1,022)**	**(773)**	**794**	**1,388**	**4,889**	**10,148**	**6,573**	**9,517**	**10,049**	**10,376**	**10,924**	**11,820**	**12,291**	**12,652**	**12,975**	**(1,537)**	**22,998**	**40,866**	**49,738**
Percent of Revenues	*-4.8%*	*-9.1%*	*-6.5%*	*6.4%*	*10.2%*	*28.3%*	*47.5%*	*28.1%*	*39.2%*	*39.5%*	*39.6%*	*40.6%*	*42.5%*	*42.6%*	*42.8%*	*42.9%*	*NM*	*30.4%*	*39.8%*	*42.7%*
Total Opex	(322)	(674)	(1,913)	(1,514)	12,838	(1,205)	(2,318)	(1,730)	(1,752)	(1,798)	(1,841)	(1,883)	(1,927)	(1,972)	(2,015)	(2,057)	(8,054)	7,585	(7,274)	(7,971)
Percent of Revenues	*2.9%*	*6.0%*	*16.2%*	*12.1%*	*-94.6%*	*7.0%*	*10.9%*	*7.4%*	*7.2%*	*7.1%*	*7.0%*	*7.0%*	*6.9%*	*6.8%*	*6.8%*	*6.8%*	*17.2%*	*-10.0%*	*7.1%*	*6.8%*
R&D	(1,354)	(1,260)	(1,297)	(1,346)	(1,615)	(1,384)	(1,404)	(1,424)	(1,300)	(1,320)	(1,340)	(1,360)	(1,380)	(1,400)	(1,420)	(1,440)	(5,257)	(5,826)	(5,320)	(5,640)
Percent of Revenues	*12.2%*	*11.2%*	*11.0%*	*10.8%*	*11.9%*	*8.0%*	*6.6%*	*6.1%*	*5.4%*	*5.2%*	*5.1%*	*5.1%*	*5.0%*	*4.9%*	*4.8%*	*4.8%*	*11.3%*	*7.7%*	*5.2%*	*4.8%*
Sales and Marketing	(116)	(111)	(113)	(122)	(170)	(150)	(222)	(243)	(243)	(254)	(262)	(269)	(278)	(288)	(296)	(303)	(463)	(785)	(1,028)	(1,165)
Percent of Revenues	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.3%*	*0.9%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*
General and Admin	(622)	(583)	(569)	(560)	(744)	(774)	(789)	(804)	(950)	(965)	(980)	(995)	(1,010)	(1,025)	(1,040)	(1,055)	(2,334)	(3,112)	(3,890)	(4,130)
Percent of Revenues	*5.6%*	*5.2%*	*4.8%*	*4.5%*	*5.5%*	*4.5%*	*3.7%*	*3.4%*	*3.9%*	*3.8%*	*3.7%*	*3.7%*	*3.6%*	*3.6%*	*3.5%*	*3.5%*	*5.0%*	*4.1%*	*3.8%*	*3.5%*
Operating Income	**(858)**	**(1,696)**	**(2,686)**	**(719)**	**14,226**	**3,684**	**7,830**	**4,843**	**7,765**	**8,250**	**8,535**	**9,041**	**9,893**	**10,319**	**10,637**	**10,918**	**(9,591)**	**30,584**	**33,592**	**41,767**
Percent of Revenues	*-7.7%*	*-15.0%*	*-22.7%*	*-5.8%*	*104.8%*	*21.3%*	*36.7%*	*20.7%*	*32.0%*	*32.4%*	*32.6%*	*33.6%*	*35.6%*	*35.8%*	*36.0%*	*36.1%*	*-20.5%*	*40.5%*	*32.7%*	*35.9%*
Total Non-operating Income(Loss)	(74)	(1,529)	(41)	127	106	204	310	204	204	204	204	204	204	204	204	204	2,114	824	816	816
Profit Before Taxes	**(932)**	**(3,226)**	**(2,727)**	**(592)**	**14,332**	**3,888**	**8,140**	**5,047**	**7,969**	**8,454**	**8,739**	**9,245**	**10,097**	**10,523**	**10,841**	**11,122**	**(7,477)**	**31,407**	**34,408**	**42,583**
Percent of Revenues	*-8.4%*	*-28.6%*	*-23.0%*	*-4.7%*	*105.6%*	*22.5%*	*38.1%*	*21.6%*	*32.8%*	*33.2%*	*33.4%*	*34.4%*	*36.3%*	*36.5%*	*36.6%*	*36.7%*	*NM*	*41.6%*	*33.5%*	*36.6%*
Change vs Year Ago	*112.9%*	*64.8%*	*-5.3%*	*-62.0%*	*-1637.2%*	*-220.5%*	*-398.5%*	*-952.2%*	*-44.4%*	*117.4%*	*7.4%*	*83.2%*	*26.7%*	*24.5%*	*24.0%*	*20.3%*				
Taxes	(165)	(108)	(1)	(62)	(101)	(597)	(1,640)	(1,009)	(1,594)	(1,691)	(1,748)	(1,849)	(2,019)	(2,105)	(2,168)	(2,224)	(336)	(3,347)	(6,882)	(8,517)
Tax Rate	*-17.7%*	*-3.3%*	*0.0%*	*-10.4%*	*0.7%*	*15.4%*	*20.1%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*-4.5%*	*10.7%*	*20.0%*	*20.0%*
Reported Income (TW GAAP)	**(1,097)**	**(3,334)**	**(2,728)**	**(654)**	**14,231**	**3,291**	**6,500**	**4,038**	**6,375**	**6,763**	**6,991**	**7,396**	**8,077**	**8,418**	**8,673**	**8,898**	**(7,813)**	**28,060**	**27,526**	**34,066**
Percent of Revenues	*-9.9%*	*-29.6%*	*-23.0%*	*-5.2%*	*104.9%*	*19.0%*	*30.5%*	*17.3%*	*26.2%*	*26.6%*	*26.7%*	*27.5%*	*29.0%*	*29.2%*	*29.3%*	*29.4%*	*-16.7%*	*37.1%*	*26.8%*	*29.2%*
Change vs Year Ago	*149.8%*	*70.3%*	*-5.2%*	*-56.4%*	*-1397.2%*	*-198.7%*	*-338.3%*	*-717.3%*	*-55.2%*	*105.5%*	*7.6%*	*83.2%*	*26.7%*	*24.5%*	*24.0%*	*20.3%*	*NM*	*NM*	*-1.9%*	*23.8%*
EPS (NT$)	**(0.26)**	**(0.80)**	**(0.65)**	**(0.16)**	**3.36**	**0.76**	**1.38**	**0.86**	**1.35**	**1.44**	**1.48**	**1.57**	**1.71**	**1.79**	**1.84**	**1.89**	**(1.87)**	**6.23**	**5.84**	**7.23**
Change vs Year Ago	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*-59.7%*	*89.6%*	*7.6%*	*83.2%*	*26.7%*	*24.5%*	*24.0%*	*20.3%*	*NM*	*NM*	*-6.2%*	*23.8%*
Weighted average shares	4,179	4,188	4,191	4,193	4,239	4,347	4,715	4,712	4,712	4,712	4,712	4,712	4,712	4,712	4,712	4,712	4,188	4,503	4,712	4,712

Source: Company data, Morgan Stanley Research (E) estimates

PSMC: Valuation Methodology

Our price target is cut to NT$54

We incorporate our revised estimates and maintain our P/B valuation methodology. Our new target, which is our base case value, is based on a 2027e P/B of 1.0x, down from 2.0x previously – below the historical average of 1.5x since 2021, to reflect pricing uncertainties.

Our base case value/price target cut of 51% is almost entirely a result of target multiple reduction, rather than change in 2027e BVPS, which declined only 1.5% in our new estimate. For PSMC, we use 1.0x P/B as the downside valuation anchor – book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns. When earnings are temporarily depressed, investors typically place greater weight on the underlying net asset base.

We decrease our bull case value from NT$177 to NT$86 and our bear case value from NT $55 to NT$27, implying 1.6x and 0.5x 2027e P/B, respectively.

We downgrade PSMC to EW

We are positive on the company's emerging 3D AI foundry business, and we believe PSMC is well positioned to benefit from its Micron partnership, including front-end migration to the one-year process node and back-end opportunities in advanced DRAM production (e.g., HBM). The company is also catching up with the DRAM pricing up-cycle.

At the same time, its logic foundry business is set to enjoy continued price hikes into 2027. However, on its memory foundry side, we see near-term pressure from DDR3/4, in which industry supply is ramping faster than demand is increasing, especially as global cloud capex may face partial delay from 2028 to 2029.

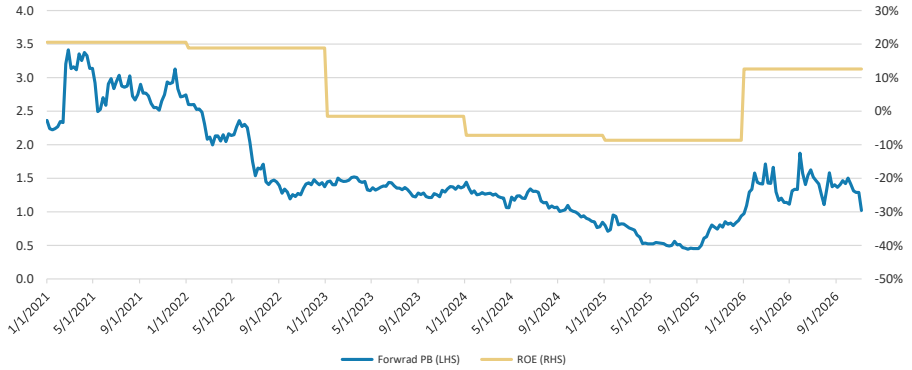
We believe the stock still has strong valuation support at 1.0x P/B, and therefore we believe an EW rating is fair.

Exhibit 26: PSMC: Historical forward P/B



Source: Company data, Morgan Stanley Research estimates

Exhibit 27: PSMC: Historical forward P/B vs. ROE



Source: Company data, Morgan Stanley Research estimates

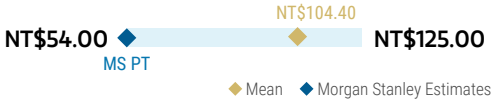
Risk Reward – Powerchip Semiconductor Manufacturing Co (6770.TW)

Pricing power uncertain, while 3D AI foundry offers upside; EW

PRICE TARGET NT$54.00

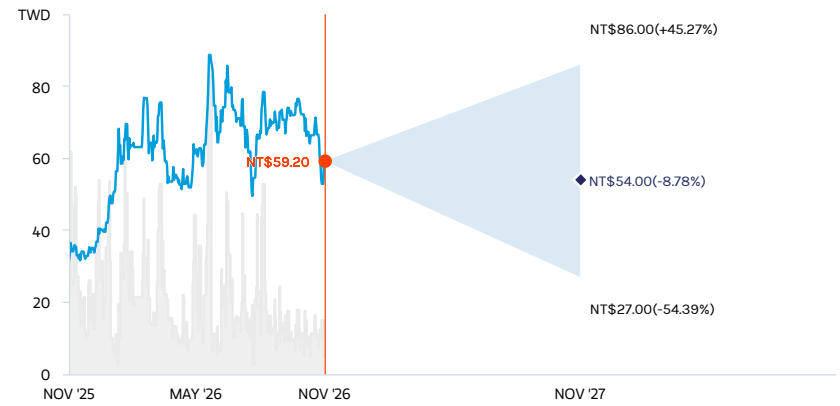
Base case, 1.0x 2027e P/B. This method is in line with our approach for Greater China memory IDM players, in view of the industry's high volatility. Our 1.0x target P/B is lower than the historical average of 1.5x since 2021, reflecting DDR3/4 pricing uncertainties. We use 1.0x P/B as the downside valuation anchor. Book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



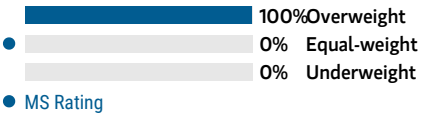
Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect a mature node capacity shortage in 2H27, reflecting fast growth of AI power ICs as a key part of future AI infrastructure.
- 3D AI foundry business is expected to be a major revenue contributor - we expect it to make up 5%, 10% and 14% of total revenue in 2026-28, respectively.
- However, DDR3/4 pricing power may be peaking .
- Our new target is based on 2027e P/B of 1.0x, below the historical average of 1.5x since 2021, reflecting DDR3/4 pricing uncertainties. We use 1.0x P/B as the downside valuation anchor - book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption:	*Positive*
Secular Growth:	*Negative*

View descriptions of Risk Rewards Themes here

BULL CASE NT$86.00

1.6x 2027e BVPS

Strong sales growth with gross margin expansion: Our assumptions include: 1) a revenue CAGR of >40% in 2025-28; 2) gross margin expanding to 45%+ by 2028; 3) mature node foundry up-cycle longer than expected; 4) faster revenue ramp-up of 3D AI foundry business; 5) specialty DRAM pricing stronger than expected, with PSMC's DRAM foundry ASP rising +30% and +5% in 2027/28 respectively.

BASE CASE NT$54.00

1.0x 2027e BVPS

Revenue growth with gross margin improvement: Our assumptions include: 1) a revenue CAGR of 36% in 2025-28; 2) a gross margin increase to 43% by 2028; 3) a mature node foundry up-cycle; 4) additional contribution from 3D AI foundry business; 5) specialty DRAM pricing stabilizing, with PSMC's DRAM foundry ASP growing +14.0% in 2027 and declining 5.4% in 2028.

BEAR CASE NT$27.00

0.5x 2027e BVPS

Revenue decline with gross margin erosion: Our assumptions include: 1) a revenue CAGR in the single digits (%) in 2025-28; 2) gross margin of <10% by 2028; 3) mature node foundry capacity no longer in shortage; 4) delays in revenue contribution from 3D AI foundry business; 5) specialty DRAM pricing decreasing sharply, with PSMC's DRAM foundry ASP rising +5% in 2027 but dropping 15% in 2028.

Risk Reward – Powerchip Semiconductor Manufacturing Co (6770.TW)

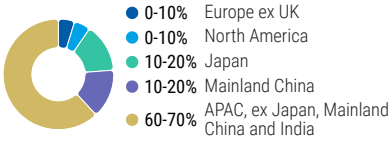
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Wafer shipment (k 8 inch)	2,614	2,881	2,965	755
Wafer ASP (USD)	457.7	474.7	517.3	522.0
Utilization rate (%) (%)	79.1	90.3	94.3	95.0

INVESTMENT DRIVERS

- Demand from consumer electronics clients
- Inventory in the supply chain
- Utilization rate
- Long-term agreements with clients

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- AI data center capacity globally holds up strongly.
- Pricing power is sustained in 2027 and 2028.
- Listing on main board attracts fund flows.
- Aluminum process enjoys higher margins.

RISKS TO DOWNSIDE

- AI data center additions face delays.
- Competition from China intensifies.
- Prices erode faster than expected.
- Demand from consumer electronics clients weakens further.
- Supply chain inventory takes longer to digest.

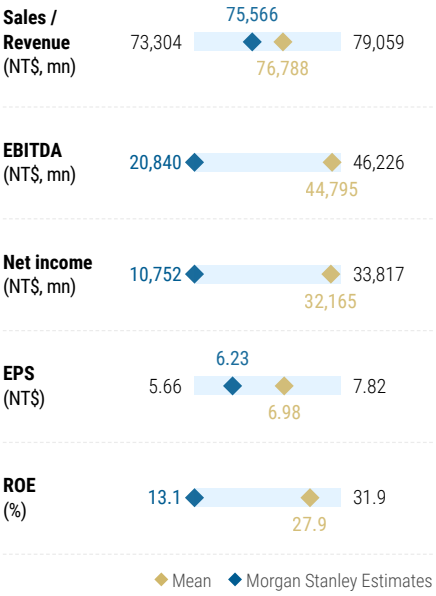
OWNERSHIP POSITIONING

Inst. Owners, % Active 25%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

PSMC: Financial Summary

Exhibit 28: Financial Summary

Income Statement NT$ mn (Years End Dec)	2025	2026e	2027e	2028e
Net sales	**46,730**	**75,566**	**102,808**	**116,499**
COGS	(48,267)	(52,568)	(61,942)	(66,761)
Gross profit	(1,537)	22,998	40,866	49,738
Operating expenses	(8,054)	7,585	(7,274)	(7,971)
R&D	(5,257)	(5,826)	(5,320)	(5,640)
SG&A	(2,797)	(3,897)	(4,918)	(5,295)
Operating income	**(9,591)**	**30,584**	**33,592**	**41,767**
Non-operating income	2,114	824	816	816
Pre-tax income	(7,477)	31,407	34,408	42,583
Income tax	(336)	(3,347)	(6,882)	(8,517)
Net income	**(7,813)**	**28,060**	**27,526**	**34,066**
Adj.wtd.avg.shrs (mn)	4,188	4,503	4,712	4,712
Reported EPS (NT$)	**(1.87)**	**6.23**	**5.84**	**7.23**
EPS for consensus (NT$)	**(1.87)**	**6.23**	**5.84**	**7.23**

Cash Flow Statement NT$mn (Years End Dec)	2025	2026e	2027e	2028e
Cash flow from Operations	**7,823**	**52,943**	**35,968**	**40,178**
Net profits	(7,813)	28,054	27,520	34,059
Depreciation	9,283	7,564	7,465	5,594
Equity investment losses (income)	11	0	0	0
Other adjustments	6,343	17,325	982	525
Cash flow from Investing	**3,554**	**41,439**	**(2,805)**	**(2,281)**
Capex	(8,591)	(15,361)	(2,805)	(2,281)
Change of LT Investment	0	0	0	0
Change of ST Investment	12,145	0	0	0
Other adjustments	(0)	56,800	0	0
Cash flow from financing	**(5,517)**	**59,944**	**23,982**	**24,142**
Increase in L/T debt	6,140	0	0	0
Increase in S/T debt	0	3,000	3,000	3,000
Cash Dividend Paid	0	2,344	(8,418)	(8,258)
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	0	29,400	29,400	29,400
Dec/Inc-TreasureSt	0	25,200	0	0
Other Adjustments	(11,657)	0	0	0
Net change in cash	**5,860**	**154,325**	**57,144**	**62,040**

Balance Sheet NT$mn (Years End Dec)	2025	2026e	2027e	2028e
Cash	16,789	171,114	228,259	290,298
Mkt Securities	2,800	2,800	2,800	2,800
AR/NR	6,550	4,830	6,571	7,446
Inventory	9,557	5,118	6,031	6,500
Other	8,451	8,451	8,451	8,451
Current Assets	**44,148**	**192,314**	**252,112**	**315,496**
Long-term investments	49	49	49	49
Fixed assets	122,004	109,520	104,861	101,547
Other assets	12,485	12,485	12,485	12,485
Total Assets	**178,686**	**314,368**	**369,506**	**429,577**
S/T borrowings	0	0	0	0
AP/NP	2,997	4,050	4,772	5,143
Other ST liabilities	19,789	32,901	38,815	43,314
LT debt	0	0	0	0
Other LT liabilities	73,900	73,900	73,900	73,900
Total Liabilities	**96,685**	**110,851**	**117,487**	**122,357**
Common shares	42,259	46,459	46,459	46,459
Retained earning	11,792	103,908	152,410	207,611
Other shareholders' equity	27,950	53,150	53,150	53,150
Total Equity	**82,000**	**203,517**	**252,019**	**307,220**
Total Liab. & Shrhldr's Equity	**178,686**	**314,368**	**369,506**	**429,577**

Financial Ratios	2025	2026e	2027e	2028e
Growth(%)				
Turnover	4.5	61.7	36.1	13.3
Operating profits	3.2	(418.9)	N/A	N/A
Pretax profits	9.4	(520.1)	9.6	23.8
Net profits	N/A	N/A	N/A	N/A
EPS	N/A	N/A	N/A	N/A
Margins (%)				
Gross Margin	(3.3)	30.4	39.8	42.7
Operating Margin	(20.5)	40.5	32.7	35.9
Pretax Margin	(16.0)	41.6	33.5	36.6
Net Profit	(16.7)	37.1	26.8	29.2
Return (%)				
ROAA	(4.3)	11.4	8.1	8.5
ROAE	(9.2)	19.7	12.1	12.2
Gearing (%)				
Net Debt/Equity	69.6	(47.8)	(61.2)	(70.4)
Liabilities/Equity	117.9	54.5	46.6	39.8
Ratios (X)				
Current ratio	1.9	5.2	5.8	6.5
Quick ratio	1.1	4.8	5.5	6.2
Others				
AR/NR Turnover (days)	23.3	23.3	23.3	23.3
Inventory Turnover (days)	35.5	35.5	35.5	35.5
AP Turnover (days)	28.1	28.1	28.1	28.1
Cash Conversion (days)	30.7	30.7	30.7	30.7

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Macronix: Earnings Estimate Revisions

We cut our 2026, 2027, and 2028 EPS forecasts 14%, 20%, and 21%, respectively: The 2026e EPS change mainly reflects actual 3Q26 results. The 2027e and 2028e EPS changes reflect a potential peak in pricing power across NOR and MLC/SLC NAND. We now reduce our NAND and /NOR wafer ASP estimates throughout 2027-2028 and 15% and 8% into 4Q28 respectively, without changing our wafer shipment estimates. We did not materially change our GM assumptions. Our changes in net income/EPS estimates are largely a function of changes in our revenue and gross profit estimates. We believe it is a good time to make these changes now as a result of the narrower shortages, moderating price hikes, and potential delay in 2028 US cloud capex.

Exhibit 29: Macronix: Earnings estimate revisions

(NT$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	91,400	99,611	-8%	169,350	198,217	-15%	170,413	200,794	-15%
Gross profit	61,873	67,438	-8%	134,841	158,581	-15%	139,838	165,449	-15%
Operating profit	47,525	54,052	-12%	115,904	141,065	-18%	120,555	147,537	-18%
Pretax income	47,939	54,767	-12%	115,876	141,791	-18%	120,588	148,299	-19%
Net income	40,499	46,345	-13%	98,122	120,082	-18%	102,113	125,594	-19%
Diluted EPS	**20.17**	**23.54**	**-14%**	**48.58**	**60.99**	**-20%**	**50.56**	**63.79**	**-21%**
Margins									
Gross margin	67.7%	67.7%	0.0 ppt	79.6%	80.0%	-0.4 ppt	82.1%	82.4%	-0.3 ppt
Operating margin	52.0%	54.3%	-2.3 ppt	68.4%	71.2%	-2.7 ppt	70.7%	73.5%	-2.7 ppt
Pretax margin	52.4%	55.0%	-2.5 ppt	68.4%	71.5%	-3.1 ppt	70.8%	73.9%	-3.1 ppt
Net margin	44.3%	46.5%	-2.2 ppt	57.9%	60.6%	-2.6 ppt	59.9%	62.5%	-2.6 ppt

NT$	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
BVPS	**41.60**	**45.11**	**-8%**	**89.66**	**106.09**	**-15%**	**140.22**	**169.88**	**-17%**

Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 30: Macronix: Quarterly financials

(NT$ mn)	1Q26	2Q26	3Q26	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Total Revenues	**10,469**	**19,125**	**24,237**	**37,570**	**39,969**	**43,806**	**42,845**	**42,730**	**42,677**	**42,626**	**42,578**	**42,533**	**28,880**	**91,400**	**169,350**	**170,413**
Sequential Change	*35.4%*	*82.7%*	*26.7%*	*55.0%*	*6.4%*	*9.6%*	*-2.2%*	*-0.3%*	*-0.1%*	*-0.1%*	*-0.1%*	*-0.1%*				
Change vs Year Ago	*70.6%*	*181.3%*	*195.1%*	*386.1%*	*281.8%*	*129.1%*	*76.8%*	*13.7%*	*6.8%*	*-2.7%*	*-0.6%*	*-0.5%*	*11.6%*	*216.5%*	*85.3%*	*0.6%*
Cost of Sales	**(6,198)**	**(6,805)**	**(7,152)**	**(9,374)**	**(8,512)**	**(8,995)**	**(8,623)**	**(8,379)**	**(7,966)**	**(7,753)**	**(7,542)**	**(7,314)**	**(23,749)**	**(29,527)**	**(34,509)**	**(30,575)**
Gross Profit	**4,271**	**12,320**	**17,086**	**28,196**	**31,457**	**34,811**	**34,222**	**34,351**	**34,711**	**34,872**	**35,035**	**35,219**	**5,131**	**61,873**	**134,841**	**139,838**
Gross Margin	40.8%	64.4%	70.5%	75.0%	78.7%	79.5%	79.9%	80.4%	81.3%	81.8%	82.3%	82.8%	17.8%	67.7%	79.6%	82.1%
Total Opex	(2,339)	(3,352)	(4,153)	(4,504)	(4,645)	(4,761)	(4,757)	(4,774)	(4,793)	(4,811)	(4,830)	(4,849)	(8,829)	(14,348)	(18,937)	(19,283)
Percent of Revenues	*22.3%*	*17.5%*	*17.1%*	*12.0%*	*11.6%*	*10.9%*	*11.1%*	*11.2%*	*11.2%*	*11.3%*	*11.3%*	*11.4%*	*30.6%*	*15.7%*	*11.2%*	*11.3%*
R&D	(1,411)	(1,870)	(2,307)	(2,327)	(2,347)	(2,367)	(2,387)	(2,407)	(2,427)	(2,447)	(2,467)	(2,487)	(5,714)	(7,915)	(9,508)	(9,828)
Percent of Revenues	*13.5%*	*9.8%*	*9.5%*	*6.2%*	*5.9%*	*5.4%*	*5.6%*	*5.6%*	*5.7%*	*5.7%*	*5.8%*	*5.8%*	*19.8%*	*8.7%*	*5.6%*	*5.8%*
Sales and Marketing	(425)	(560)	(601)	(931)	(991)	(1,086)	(1,062)	(1,059)	(1,058)	(1,057)	(1,056)	(1,054)	(1,520)	(2,517)	(4,199)	(4,225)
Percent of Revenues	*4.1%*	*2.9%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*5.3%*	*2.8%*	*2.5%*	*2.5%*
General and Admin	(503)	(922)	(1,245)	(1,245)	(1,308)	(1,308)	(1,308)	(1,308)	(1,308)	(1,308)	(1,308)	(1,308)	(1,594)	(3,916)	(5,230)	(5,230)
Percent of Revenues	*4.8%*	*4.8%*	*5.1%*	*3.3%*	*3.3%*	*3.0%*	*3.1%*	*3.1%*	*3.1%*	*3.1%*	*3.1%*	*3.1%*	*5.5%*	*4.3%*	*3.1%*	*3.1%*
Operating Income	**1,933**	**8,968**	**12,932**	**23,692**	**26,811**	**30,051**	**29,465**	**29,577**	**29,919**	**30,061**	**30,205**	**30,370**	**(3,698)**	**47,525**	**115,904**	**120,555**
Percent of Revenues	*18.5%*	*46.9%*	*53.4%*	*63.1%*	*67.1%*	*68.6%*	*68.8%*	*69.2%*	*70.1%*	*70.5%*	*70.9%*	*71.4%*	*-12.8%*	*52.0%*	*68.4%*	*70.7%*
Total Non-operating Income(Loss)	**134**	**232**	**65**	**(17)**	**(13)**	**(9)**	**(5)**	**(1)**	**5**	**9**	**12**	**8**	**64**	**414**	**(28)**	**33**
Profit Before Taxes	**2,066**	**9,200**	**12,997**	**23,675**	**26,798**	**30,042**	**29,460**	**29,577**	**29,923**	**30,070**	**30,217**	**30,378**	**(3,633)**	**47,939**	**115,876**	**120,588**
Percent of Revenues	*19.7%*	*48.1%*	*53.6%*	*63.0%*	*67.0%*	*68.6%*	*68.8%*	*69.2%*	*70.1%*	*70.5%*	*71.0%*	*71.4%*	*-12.6%*	*52.4%*	*68.4%*	*70.8%*
Change vs Year Ago	*-313.2%*	*-753.6%*	*-1462.6%*	*-7927.8%*	*1196.8%*	*226.5%*	*126.7%*	*24.9%*	*11.7%*	*0.1%*	*2.6%*	*2.7%*				
Taxes	(287)	(1,464)	(2,046)	(3,622)	(4,100)	(4,596)	(4,507)	(4,525)	(4,578)	(4,601)	(4,623)	(4,648)	328	(7,420)	(17,729)	(18,450)
Tax Rate	*-13.9%*	*-15.9%*	*-15.7%*	*15.3%*	*15.3%*	*15.3%*	*15.3%*	*15.3%*	*15.3%*	*15.3%*	*15.3%*	*15.3%*	*9.0%*	*15.5%*	*15.3%*	*15.3%*
Net Income, Cont Ops	1,779	7,736	10,951	20,053	22,698	25,445	24,953	25,051	25,345	25,470	25,594	25,730	(3,305)	40,519	98,147	102,138
Percent of Revenues	*17.0%*	*40.4%*	*45.2%*	*53.4%*	*56.8%*	*58.1%*	*58.2%*	*58.6%*	*59.4%*	*59.8%*	*60.1%*	*60.5%*	*-11.4%*	*44.3%*	*58.0%*	*59.9%*
Minority Interest	(4)	(4)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(3)	(20)	(25)	(25)
Reported Income (TW GAAP)	**1,776**	**7,732**	**10,945**	**20,047**	**22,691**	**25,439**	**24,946**	**25,045**	**25,339**	**25,463**	**25,588**	**25,724**	**(3,308)**	**40,499**	**98,122**	**102,113**
Percent of Revenues	*17.0%*	*40.4%*	*45.2%*	*53.4%*	*56.8%*	*58.1%*	*58.2%*	*58.6%*	*59.4%*	*59.7%*	*60.1%*	*60.5%*	*-11.5%*	*44.3%*	*57.9%*	*59.9%*
Change vs Year Ago	*-303.4%*	*-705.9%*	*-1367.8%*	*-6890.0%*	*1177.8%*	*229.0%*	*127.9%*	*24.9%*	*11.7%*	*0.1%*	*2.6%*	*2.7%*	*2.9%*	*-1324.4%*	*142.3%*	*4.1%*
Reported Diluted EPS (NT$, TW GAAP)	**0.90**	**3.93**	**5.42**	**9.93**	**11.23**	**12.60**	**12.35**	**12.40**	**12.55**	**12.61**	**12.67**	**12.74**	**(1.78)**	**20.17**	**48.58**	**50.56**
Reported Basic EPS (NT$, GAAP)	**0.90**	**3.93**	**5.52**	**10.11**	**11.44**	**12.83**	**12.58**	**12.63**	**12.78**	**12.84**	**12.90**	**12.97**	**(1.78)**	**20.50**	**49.48**	**51.50**

Source: Company data, Morgan Stanley Research (E) estimates

Macronix: Valuation Methodology

We cut our price target to NT$90 from NT$220

We continue to use a P/B multiple methodology to derive our price target, which is our base case value, in line with our approach for Greater China memory IDM peers and in view of the industry's high volatility. Our new target is based on a 2027e P/B of 1.0x, down from 2.1x previously – below the average of 1.5x since 2017, given the potential peak in pricing power for NOR and SLC/MLC NAND. For Macronix, we now use 1.0x P/B as the downside valuation anchor because book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns. When earnings are temporarily depressed, investors typically place greater weight on the underlying net asset base.

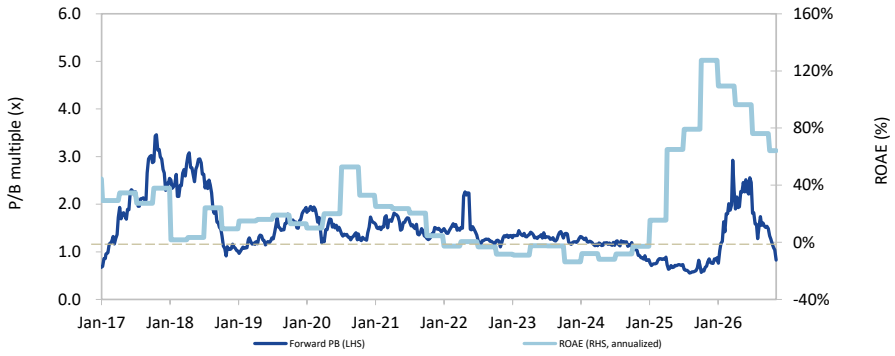
Our base case value/price target cut of 59% reflects both target multiple reduction and a 15% 2027 BVPS estimate reduction (relatively large because of a smaller base).

We cut our bull case value from NT$355 to NT$145, which is 1.6x our 2027e BVPS. We cut our bear case value from NT$100 to NT$41, which is 0.5x our 2027e BVPS.

We downgrade Macronix to EW

We continue to like its unique position in MLC NAND as the only major supplier still ramping up capacity. However, we do see near-term pressure from NOR, DDR3/4, and even SLC/MLC NAND, in all of which industry supply is ramping faster than demand is increasing, especially as global cloud capex may face partial delay from 2028 to 2029. We believe the stock still has strong valuation support at 1.0x P/B, and therefore we believe an EW rating is fair.

Exhibit 31: Forward P/B vs. ROAE



Source: Factset, Morgan Stanley Research

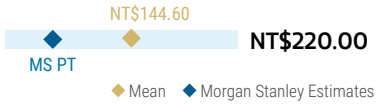
Risk Reward – Macronix International Co Ltd (2337.TW)

Pricing power may be peaking; EW

PRICE TARGET NT$90.00

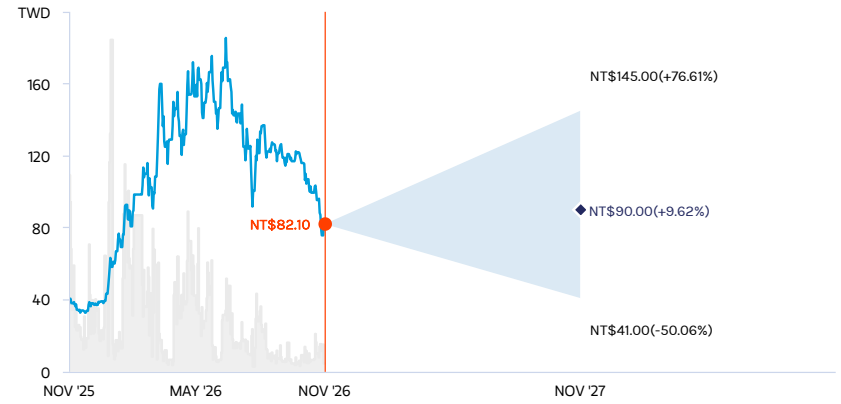
Base case, P/B, in line with our Greater China memory coverage, given the industry's cyclical nature. We apply a target multiple of 1.0x to our 2027 BVPS estimate vs. its historical average of 1.5x since 2017, given the potential peak in pricing power of NOR and SLC/MLC NAND. We use 1.0x P/B as the downside valuation anchor. Book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



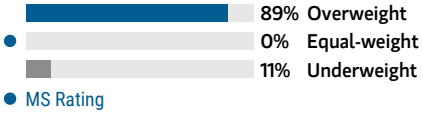
Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Rising uncertainty over cloud spending weakens confidence in incremental enterprise demand, while much weaker consumer demand challenges customers' ability to absorb further price hikes
- Capacity expansion captures market upside but may create utilization or margin issues should pricing peak.
- Our price target is set at 1.0x 2027e P/B, vs the average of 1.5x since 2017, given the potential peak in pricing power of NOR and SLC/MLC NAND. For Macronix, we now use 1.0x P/B as the downside valuation anchor - book value provides a psychological support level for capital-intensive IDM/foundry stocks during cyclical downturns.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Negative*

View descriptions of Risk Rewards Themes here

BULL CASE NT$145.00

1.6x 2027e BVPS

Pricing power remains stronger for longer: NAND wafer ASP rises 40% and 20% in 2027 and 2028, while NOR wafer ASP rises 40% and 20% in 2027/28.

BASE CASE NT$90.00

1.0x 2027e BVPS

Pricing power may be peaking into 2027: NAND wafer ASP rises 18% and 0% in 2027 and 2028, while NOR wafer ASP rises 25% and 0% in 2027 and 2028.

BEAR CASE NT$41.00

0.5x 2027e BVPS

Pricing significantly declines beyond 2027: NAND wafer ASP grows 10% in 2027 but drops 10% in 2028, while NOR wafer ASP rises 10% in 2027 but drops 110% in 2028.

Risk Reward – Macronix International Co Ltd (2337.TW)

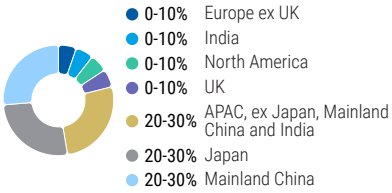
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
NOR Flash pricing (NT$)	1,721	3,501	4,361	4,361
ROM sales (NT$, mn)	5,898	3,471	406	0
NAND sales (NT$, mn)	3,540	42,858	127,678	135,587

INVESTMENT DRIVERS

- NOR pricing trend
- SLC NAND pricing
- Peers' production expansion

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Wider legacy NAND supply/demand gap
- NOR up-cycle, driven by limited supply expansion and/or stronger demand
- Faster growth for Switch hardware and core titles
- Healthy utilization; sustained margins

RISKS TO DOWNSIDE

- Less severe legacy NAND supply/demand gap
- NOR down-cycle, driven by faster supply expansion and/or demand weakness
- Slower growth for Switch hardware and core titles
- Weak utilization; margin erosion

OWNERSHIP POSITIONING

Inst. Owners, % Active 34.8%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	76,334	91,400 / 85,691	99,611
EBITDA (NT$, mn)	38,309	54,041 / 47,316	61,530
Net income (NT$, mn)	29,195	40,499 / 36,147	46,345
EPS (NT$)	14.76	20.17 / 18.28	23.54

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Macronix: Financial Summary

Exhibit 32: Financial Summary

Income Statement, 2025-2028e, Year End Dec

(NT$ mn)	2025	2026e	2027e	2028e
Turnover	**28,880**	**91,400**	**169,350**	**170,413**
YoY Growth	*11.6%*	*216.5%*	*85.3%*	*0.6%*
Less: COGS	**(23,749)**	**(29,527)**	**(34,509)**	**(30,575)**
Variable costs	(19,503)	(24,051)	(29,195)	(25,989)
Depreciation & amort	(4,246)	(5,476)	(5,314)	(4,586)
Gross profit	**5,131**	**61,873**	**134,841**	**139,838**
YoY Growth	*-15.9%*	*1105.9%*	*117.9%*	*3.7%*
% margin	*17.8%*	*67.7%*	*79.6%*	*82.1%*
Operating Expenses:	**(8,829)**	**(14,348)**	**(18,937)**	**(19,283)**
R&D	(5,714)	(7,915)	(9,508)	(9,828)
Sales and Marketing	(1,520)	(2,517)	(4,199)	(4,225)
General and Admin	(1,594)	(3,916)	(5,230)	(5,230)
Operating Profit	**(3,698)**	**47,525**	**115,904**	**120,555**
YoY Growth	*-5.8%*	*-1385.3%*	*143.9%*	*4.0%*
% margin	*-12.8%*	*52.0%*	*68.4%*	*70.7%*
Total Non-op	**64**	**414**	**(28)**	**33**
Pretax Profit	(3,633)	47,939	115,876	120,588
YoY Growth	*2.5%*	*-1419.4%*	*141.7%*	*4.1%*
% margin	*-12.6%*	*52.4%*	*68.4%*	*70.8%*
Tax	328	(7,420)	(17,729)	(18,450)
Reported Net Income	(3,308)	40,499	98,122	102,113
Reported EPS (NT$)	**(1.78)**	**20.17**	**48.58**	**50.56**
ModelWare EPS (NT$)	**(1.78)**	**20.17**	**48.58**	**50.56**

Key Ratios

	2025	2026e	2027e	2028e
Return (%)				
ROAA	*(0.0)*	*0.2*	*0.4*	*0.3*
ROAE	*(0.1)*	*0.3*	*0.5*	*0.4*
OP. ATO	*46.7%*	*78.6%*	*135.8%*	*143.3%*
Gearing (x)				
Net Debt/ Equity	0.1x	0.1x	(0.5x)	(0.7x)
Current Ratio	2.4x	5.9x	14.1x	22.7x
Quick Ratio	1.6x	4.2x	12.6x	21.4x
Operating Cycle				
AR/NR Turnover (days)	129	79	54	56
Inventory Turnover (days)	232	215	211	224
AP Turnover (days)	55	50	46	49
Cash Conversion (days)	305	244	219	231

e = Morgan Stanley Research Estimates
Source: Company Data, Morgan Stanley Research

Balance sheet

(NT$ mn)	2025	2026e	2027e	2028e
Assets				
Cash & Equivalents	14,913	36,971	146,435	254,465
Marketable Security	7	7	7	7
A/R & N/R	4,315	23,290	26,489	26,366
Inventories	9,813	22,714	20,304	17,723
Other Current Asset	190	190	190	190
Total current assets	**29,425**	**83,359**	**193,612**	**298,938**
LT Investment	6,618	6,614	6,610	6,608
Total Fixed Assets	38,549	53,619	48,966	45,192
Total Other Assets	3,909	3,909	3,909	3,909
Total Assets	**78,500**	**147,501**	**253,097**	**354,647**
A/P & N/P	3,120	4,958	4,432	3,868
Accrued Expenses	0	0	0	0
Other Payable	1,877	1,877	1,877	1,877
Total Current Liab.	**12,404**	**14,241**	**13,715**	**13,152**
L-T Liabilities	17,386	47,386	55,386	55,386
Total Other L-T Liab	2,903	2,903	2,903	2,903
Total Liabilities	**32,693**	**64,530**	**72,004**	**71,441**
Common Stock	18,573	18,573	18,573	18,573
Preferred Stock				
Capital Reserve	1,472	1,472	1,472	1,472
Retained earnings	25,916	63,079	161,201	263,315
Treasury Stock	(159)	(159)	(159)	(159)
Total Equity	**45,808**	**82,970**	**181,093**	**283,206**

Cash Flow Statement

(NT$ mn)	2025	2026e	2027e	2028e
Net Income	(3,308)	40,499	98,122	102,113
Depreciation	5,094	6,517	6,411	5,532
Net Investment Losses (Gains)	(2)	0	0	0
Others	3,051	(30,035)	(1,311)	2,143
Cash Flow-Operating	**4,835**	**16,982**	**103,222**	**109,788**
(Purchase) of FA	(3,000)	(21,587)	(1,758)	(1,758)
Sale of Fix Asset	2	0	0	0
(Purchase)L-T Inv.	0	0	0	0
Sale of L-T Inv.	0	0	0	0
Sale(Pur.)S-T Inv.	144	0	0	0
Cash Flow-Investing	**(2,884)**	**(21,587)**	**(1,758)**	**(1,758)**
Inc(Dec)-S-T Debt	(2,488)	0	0	0
Dividend Paid	(3,337)	(3,337)	(3,337)	(3,337)
Dir.&Emp.Bonus	0	0	0	0
Proceed from New Issue	0	0	0	0
Dec/Inc-TreasureSt	0	0	0	0
Others	2,884	30,000	8,000	0
Cash Flow-Financing	**(2,941)**	**26,663**	**4,663**	**(3,337)**
Change in Cash	**3,289**	**22,058**	**106,127**	**104,693**
Net cash/(debt), b/f	57,530	60,819	82,877	189,005
Net cash/(debt), c/f	**60,819**	**82,877**	**189,005**	**293,698**

Nanya Tech: Estimate Revisions and Quarterly Financials

Our 2026 estimates are unchanged; we cut our 2027 and 2028 EPS estimates 7% and 9%,respectively: We factor in a potential peak in DDR3/4 pricing power, which affects Nanya Tech's top line and margins. We now reduce our Nanya Tech DRAM ASP per bit estimates throughout 2027-2028 and 6% into 4Q28, without changing our wafer shipment estimates. We did not materially change our GM assumptions. Our changes in net income/EPS estimates are largely a function of changes in our revenue and gross profit estimates. We believe it is a good time to make these changes now as a result of the narrower shortages, moderating price hikes, and potential delay in 2028 US cloud capex.

Exhibit 33: Nanya Tech: Estimate revisions

NT$ mn	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
Net sales	405,725	405,725	0%	688,388	728,810	-6%	550,151	585,384	-6%
Gross profit	333,069	333,069	0%	569,281	609,702	-7%	391,494	426,727	-8%
Operating profit	310,209	310,209	0%	520,352	559,124	-7%	340,542	374,295	-9%
Pretax Income	316,087	316,087	0%	521,872	560,644	-7%	342,074	375,827	-9%
Net income	257,292	257,292	0%	424,452	455,960	-7%	278,413	305,892	-9%
EPS (NT$)	**75.96**	**75.96**	**0%**	**123.03**	**132.17**	**-7%**	**80.70**	**88.67**	**-9%**
Margins									
Gross margin	82.1%	82.1%		82.7%	83.7%		71.2%	72.9%	
Operating margin	76.5%	76.5%		75.6%	76.7%		61.9%	63.9%	
Pretax margin	77.9%	77.9%		75.8%	76.9%		62.2%	64.2%	
Net margin	63.4%	63.4%		61.7%	62.6%		50.6%	52.3%	

NT$	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
BVPS (NT$)	146.00	146.00	0%	269.02	278.16	-3%	349.71	366.81	-5%

Source: Morgan Stanley Research (E) estimates

Exhibit 34: Nanya Tech: Quarterly financials

NT$ in million	1Q26	2Q26	3Q26P	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026e	2027e	2028e
Total Revenues	**49,087**	**82,549**	**133,649**	**140,440**	**165,484**	**178,632**	**179,294**	**164,977**	**34,132**	**66,587**	**405,725**	**688,388**	**550,151**
Sequential Change	*63.1%*	*68.2%*	*61.9%*	*5.1%*	*17.8%*	*7.9%*	*0.4%*	*-8.0%*					
Change vs Year Ago	*582.9%*	*684.2%*	*611.7%*	*366.7%*	*237.1%*	*116.4%*	*34.2%*	*17.5%*	*14.2%*	*95.1%*	*509.3%*	*69.7%*	*-20.1%*
Cost of Sales	(15,771)	(16,930)	(19,167)	(20,788)	(22,805)	(25,384)	(28,086)	(42,833)	(32,292)	(50,841)	(72,656)	(119,107)	(158,657)
Percent of Revenues	*32.1%*	*20.5%*	*14.3%*	*14.8%*	*13.8%*	*14.2%*	*15.7%*	*26.0%*	*94.6%*	*76.4%*	*17.9%*	*17.3%*	*28.8%*
Variable costs	(12,874)	(13,970)	(16,102)	(17,449)	(19,924)	(22,502)	(25,204)	(28,007)	(18,408)	(37,321)	(60,394)	(95,637)	(133,967)
as a % of revenue	*26.2%*	*16.9%*	*12.0%*	*12.4%*	*12.0%*	*12.6%*	*14.1%*	*17.0%*	*53.9%*	*56.0%*	*14.9%*	*13.9%*	*24.4%*
Dep. & Amort. expense	(2,897)	(2,960)	(3,065)	(3,340)	(2,882)	(2,882)	(2,882)	(14,825)	(13,883)	(13,520)	(12,262)	(23,470)	(24,690)
as a % of COGS	*18.4%*	*17.5%*	*16.0%*	*16.1%*	*12.6%*	*11.4%*	*10.3%*	*34.6%*	*43.0%*	*26.6%*	*16.9%*	*19.7%*	*15.6%*
as a % of revenue	*5.9%*	*3.6%*	*2.3%*	*2.4%*	*1.7%*	*1.6%*	*1.6%*	*9.0%*	*40.7%*	*20.3%*	*3.0%*	*3.4%*	*4.5%*
Gross Margin	**33,316**	**65,619**	**114,482**	**119,652**	**142,679**	**153,249**	**151,208**	**122,145**	**1,840**	**15,745**	**333,069**	**569,281**	**391,494**
Percent of Revenues	*67.9%*	*79.5%*	*85.7%*	*85.2%*	*86.2%*	*85.8%*	*84.3%*	*74.0%*	*5.4%*	*23.6%*	*82.1%*	*82.7%*	*71.2%*
Total Opex	(3,205)	(4,793)	(7,588)	(7,275)	(11,119)	(12,095)	(12,796)	(12,919)	(10,134)	(9,741)	(22,860)	(48,929)	(50,952)
Percent of Revenues	*6.5%*	*5.8%*	*5.7%*	*5.2%*	*6.7%*	*6.8%*	*7.1%*	*7.8%*	*29.7%*	*14.6%*	*5.6%*	*7.1%*	*9.3%*
R&D	(2,109)	(3,230)	(5,145)	(4,334)	(4,500)	(4,950)	(5,445)	(5,990)	(7,685)	(7,041)	(14,818)	(20,885)	(27,846)
Percent of Revenues	*4.3%*	*3.9%*	*3.8%*	*3.1%*	*2.7%*	*2.8%*	*3.0%*	*3.6%*	*22.5%*	*10.6%*	*3.7%*	*3.0%*	*5.1%*
Sales and Marketing	(347)	(503)	(594)	(856)	(1,655)	(1,786)	(1,793)	(1,650)	(665)	(832)	(2,301)	(6,884)	(6,602)
Percent of Revenues	*0.7%*	*0.6%*	*0.4%*	*0.6%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.9%*	*1.2%*	*0.6%*	*1.0%*	*1.2%*
General and Admin	(748)	(1,060)	(1,849)	(2,084)	(4,965)	(5,359)	(5,558)	(5,279)	(1,784)	(1,868)	(5,742)	(21,161)	(16,505)
Percent of Revenues	*1.5%*	*1.3%*	*1.4%*	*1.5%*	*3.0%*	*3.0%*	*3.1%*	*3.2%*	*5.2%*	*2.8%*	*1.4%*	*3.1%*	*3.0%*
Operating Income	**30,111**	**60,826**	**106,894**	**112,377**	**131,560**	**141,153**	**138,412**	**109,226**	**(8,294)**	**6,004**	**310,209**	**520,352**	**340,542**
Percent of Revenues	*61.3%*	*73.7%*	*80.0%*	*80.0%*	*79.5%*	*79.0%*	*77.2%*	*66.2%*	*-24.3%*	*9.0%*	*76.5%*	*75.6%*	*61.9%*
EBITDA	33,008	63,786	109,959	115,717	134,441	144,035	141,294	124,052	5,589	19,524	322,470	543,822	365,232
EBITDA margin	*67%*	*77%*	*82%*	*82%*	*81%*	*81%*	*79%*	*75%*	*16%*	*29%*	*79%*	*79%*	*66%*
Non-operating Income(Loss)	**1,607**	**1,503**	**2,438**	**330**	**381**	**381**	**381**	**378**	**3,998**	**2,639**	**5,878**	**1,520**	**1,532**
Profit Before Taxes	**31,718**	**62,329**	**109,332**	**112,707**	**131,941**	**141,534**	**138,793**	**109,604**	**(4,297)**	**8,643**	**316,087**	**521,872**	**342,074**
Percent of Revenues	*64.6%*	*75.5%*	*81.8%*	*80.3%*	*79.7%*	*79.2%*	*77.4%*	*66.4%*	*-12.6%*	*13.0%*	*77.9%*	*75.8%*	*62.2%*
Taxes	(5,660)	(12,138)	(21,705)	(19,292)	(23,544)	(27,561)	(27,554)	(18,761)	1,474	(1,269)	(58,795)	(97,420)	(63,661)
Tax Rate	*17.8%*	*19.5%*	*19.9%*	*17.1%*	*17.8%*	*19.5%*	*19.9%*	*17.1%*	*34.3%*	*14.7%*	*18.6%*	*18.7%*	*18.6%*
Reported Income (TW GAAP)	**26,058**	**50,192**	**87,627**	**93,415**	**108,397**	**113,973**	**111,239**	**90,843**	**(2,823)**	**7,373**	**257,292**	**424,452**	**278,413**
Percent of Revenues	*53.1%*	*60.8%*	*65.6%*	*66.5%*	*65.5%*	*63.8%*	*62.0%*	*55.1%*	*-8.3%*	*11.1%*	*63.4%*	*61.7%*	*50.6%*
Change vs Year Ago	*-1443%*	*-1324%*	*5505%*	*742%*	*316%*	*127%*	*27%*	*-3%*	*-59%*	*-361%*	*3389%*	*65%*	*-34%*
Reported EPS (NT$, TW GAAP)	**8.08**	**14.66**	**25.40**	**27.08**	**31.42**	**33.04**	**32.24**	**26.33**	**(0.91)**	**2.36**	**75.96**	**123.03**	**80.70**
Sequential Change	*129%*	*81%*	*73%*	*7%*	*16%*	*5%*	*-2%*	*-18%*	*-59%*	*-359%*	*3117%*	*62%*	*-34%*
Book value per share									**53.3**	**54.6**	**148.7**	**269.1**	**349.7**

Source: Company data, Morgan Stanley Research (E) estimates

Nanya Tech: Valuation Methodology

We cut our price target from NT$600 to NT$550

Our price target (base case scenario value) reflects a target BVPS multiple of 2.04x for 2027 (was 4.47x) versus ROE of 59% for 2027. We lower our target multiple to reflect potential peak in DDR4 pricing power and recent memory segment volatility. For Nanya Tech, our new base case target P/B multiple of 2.04x P/B is stronger than our downside valuation anchor of 1.0x P/B for capital-intensive IDM/foundry stocks. This is because we expect Nanya to remain relatively resilient compared to other DDR4 players in our coverage, given stronger process technology and potential DDR5 migration.

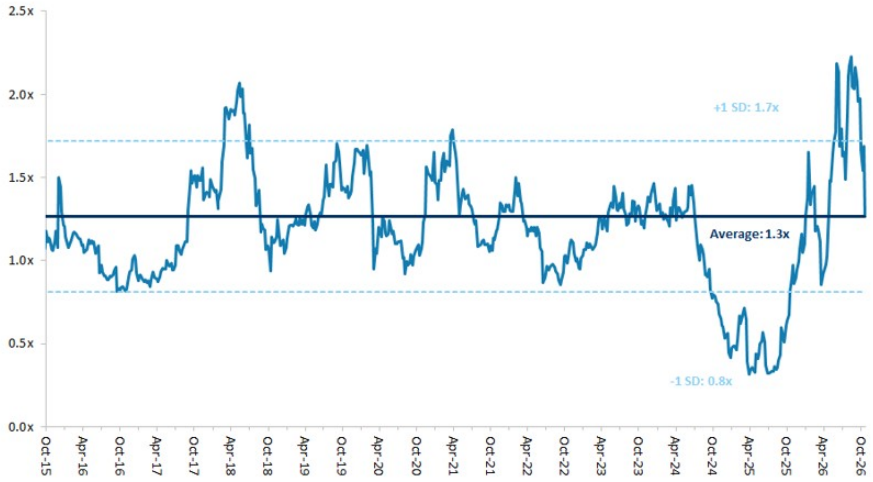
Our base case value/price target cut of 8% reflects both target multiple reduction and BVPS estimate adjustment of 3% arising from earnings estimate revisions.

We cut our bull and bear case values from NT$1,270 and NT$330 to NT$1,150 and NT $300, respectively -- 4.27x and 1.12x 2027e BVPS.

We remain OW

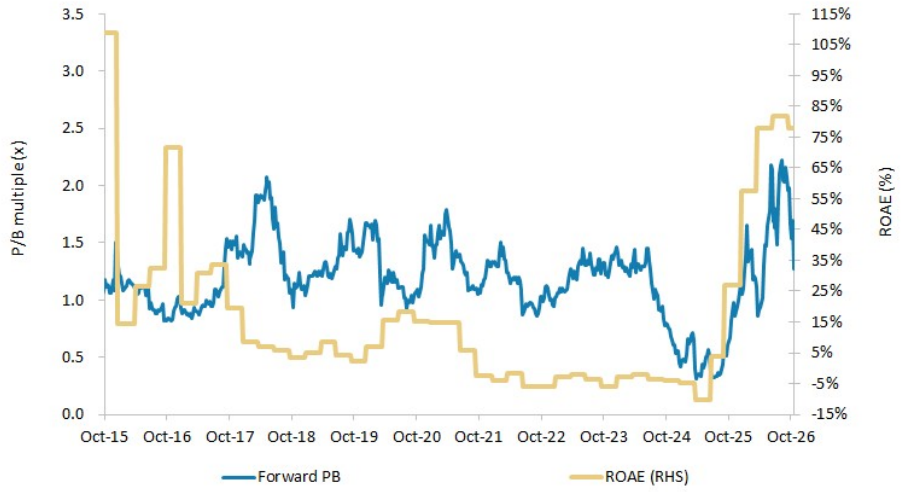
Despite potentially peaking pricing power seen in DDR4, we remain positive on DDR5 in view of continued strong demand from servers. Nanya is already producing DDR5 on its second-generation 10-nm process, with DDR5 contributing ~10% of revenue; its next process enters customer qualification in 1Q27, and EUV implementation is targeted for 2028. This establishes its edge in process technology vs the other DDR4 vendors in our coverage and a stronger position in DDR5 migration. Valuation is also attractive, in our view. Therefore, we remain OW on Nanya Tech.

Exhibit 35: Nanya Tech – Historical P/B band



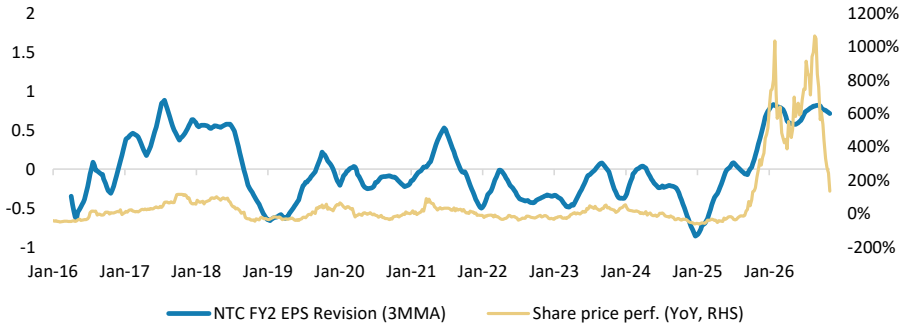
Source: Company data, Morgan Stanley Research estimates

Exhibit 36: Nanya Tech: Historical forward P/B vs. ROE



Source: Company data, Morgan Stanley Research estimates

Exhibit 37: Nanya Tech: Earnings estimate revision breadth



Source: Company data, Morgan Stanley Research estimates

Nanya Tech: Financial Summary

Income Statement

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	66,587	405,725	688,388	550,151
COGS	(50,841)	(72,656)	(119,107)	(158,657)
Depreciation Expense	(13,520)	(12,262)	(23,470)	(24,690)
Variable Cost	(37,321)	(60,394)	(95,637)	(133,967)
Gross profit	15,745	333,069	569,281	391,494
Operating expenses	(9,741)	(22,860)	(48,929)	(50,952)
Operating income	**6,004**	**310,209**	**520,352**	**340,542**
Non-operating income	2,639	5,878	1,520	1,532
Pre-tax income	8,643	316,087	521,872	342,074
Income tax	(1,269)	(58,795)	(97,420)	(63,661)
Reported net Income	**7,373**	**257,292**	**424,452**	**278,413**
Adj.wtd.avg.shrs(m)	3,123	3,387	3,450	3,450
Reported EPS (NT$)	**2.36**	**75.96**	**123.03**	**80.70**
EPS for consensus (NT$)	**2.36**	**75.96**	**123.03**	**80.70**

Balance Sheet

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	58,074	468,374	855,763	1,135,449
Mkt Securities	0	0	0	0
AR/NR	16,558	71,688	84,213	60,714
Inventory	27,288	26,437	54,472	56,422
Other	6,621	20,033	22,905	17,516
Current Assets	108,541	586,532	1,017,353	1,270,102
Long-term investments	5,904	8,476	9,415	10,354
Fixed assets	85,031	128,544	135,074	140,384
Other assets	8,977	5,426	5,426	5,426
Total Assets	**208,453**	**728,978**	**1,167,267**	**1,426,265**
S/T borrowings	5,060	160,000	165,000	145,000
AP/NP	6,481	8,308	17,117	17,730
Other ST liabilities	7,480	33,045	33,074	33,076
LT debt	14,196	18,397	18,397	18,397
Other LT liabilities	4,698	5,488	5,488	5,488
Total Liabilities	**37,914**	**225,237**	**239,076**	**219,690**
Common stocks	30,986	34,502	34,502	34,502
Preferred stocks	0	0	0	0
Capital Reearve	33,081	108,459	108,459	108,459
Retained earning	106,333	360,642	785,093	1,063,476
Other shareholders' equity	0	0	0	0
Total Equity	170,538	503,741	928,192	1,206,574
Total Liab. & Shrhldr's Equity	**208,453**	**728,978**	**1,167,267**	**1,426,265**
BVPS	**54.61**	**148.72**	**269.05**	**349.74**

Cash Flow Statement

NT$mn (Years End	2025	2026E	2027E	2028E
Cashflow from C	**18,568**	**223,324**	**412,390**	**329,716**
Net profits	7,373	257,292	424,452	278,413
Depreciation	14,034	12,262	23,470	24,690
Working Capital C	5,697	(52,453)	(31,750)	22,162
Other adjustment	(8,536)	6,223	(3,782)	4,451
Cashflow from I	**(14,110)**	**(52,201)**	**(30,000)**	**(30,000)**
Capex	(13,441)	(51,996)	(30,000)	(30,000)
Change of LT Inv	(612)	0	0	0
Change of ST Inv	0	0	0	0
Other adjustment	(57)	(206)	0	0
Cashflow from f	**(6,317)**	**237,868**	**4,999**	**(20,030)**
Increase in L/T de	(6,750)	4,201	0	0
Increase in S/T d	(9,477)	154,941	5,000	(20,000)
Cash Dividend Pa	0	0	(1)	(30)
Dir& Emp Bonus	(7,000)	0	0	0
Proceed fmNewIs	0	78,718	0	0
Other adjustment	16,909	9	0	0
Exchange rate ac	(1,970)	1,309	0	(0)
Net change in ca	**-3,829**	**410,300**	**387,389**	**279,686**

Financial Ratios

	2025	2026E	2027E	2028E
Growth(%)				
Turnover	95.1	509.3	69.7	-20.1
Margins (%)				
Gross Margin	23.6	82.1	82.7	71.2
Operating Margin	9.0	76.5	75.6	61.9
EBITA Margin	-11.3	73.4	72.2	57.4
Pretax Margin	13.0	77.9	75.8	62.2
Net Profit	11.1	63.4	61.7	50.6
Return (%)				
ROAE	4.4	76.3	59.3	26.1
ROAA	3.6	54.9	44.8	21.5
Gearing (%)				
Net Debt/Equity	(22.7)	(57.6)	(72.4)	(80.6)
Liabilities/Equity	22.2	44.7	25.8	18.2
Ratios (X)				
Current ratio	5.7	2.9	4.7	6.5
Quick ratio	3.9	2.7	4.4	6.1
Others				
AR/NR Turnover	50	57	40	41
Inventory Turnov	337	172	24	21
AP Turnover (day	59	47	42	52
Cash Conversion	328	182	22	10

Source: Company data, TEJ, Morgan Stanley ResearchE= Morgan Stanley Research Estimates

Risk Reward – Nanya Technology Corp. (2408.TW)

Still resilient during a DDR4 pricing power debate; OW

PRICE TARGET NT$550.00

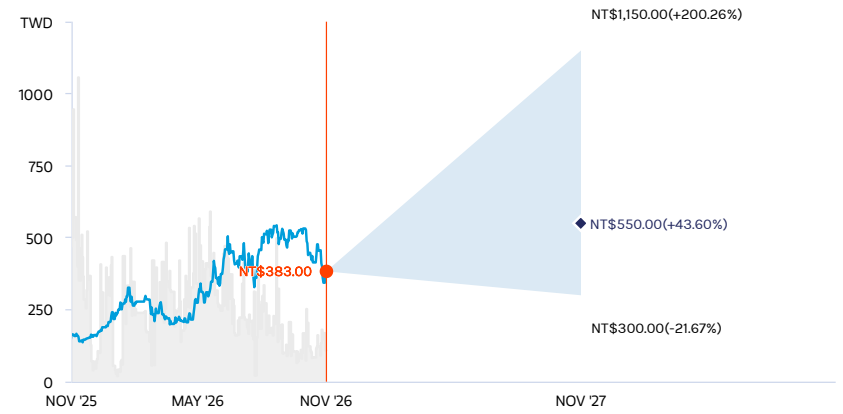
Base case, P/B. We expect the stock to trade at 2.04x 2027e BVPS, justified by 2027e ROE of 59%. We think NTC is well positioned compared to other DDR4 players given stronger process technology and the DDR5 migration. Thus, we ascribe a multiple stronger than our downside valuation anchor of 1.0x P/B for capital-intensive IDM/foundry stocks.

Consensus Price Target Distribution NT$150.00 — NT$632.64 — MS PT — NT$800.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Major memory suppliers are gradually exiting the DDR4 market, benefiting existing Taiwanese players, including Nanya Tech.
- We factor in a potential peak in DDR4 pricing power. However, we expect Nanya to remain relatively resilient compared to other DDR4 players in our coverage, given stronger process technology and potential DDR5 migration.
- Thus, we set our price target at P/B of 2.04x for 2027e, stronger than our downside valuation anchor of 1.0x P/B for capital-intensive IDM/foundry stocks.

Consensus Rating Distribution

93%	Overweight
0%	Equal-weight
7%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power:	*Positive*
Secular Growth:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$1,150.00

4.3x 2027e BVPS

DDR4 pricing continues to rise: Supply ramp is slower than expected. 1a/1b ramp and DDR5 migration are faster than expected.

BASE CASE NT$550.00

2.0x 2027e BVPS

DDR4 pricing remains stable: Supply ramp is offset by demand improvement. 1a/1b ramp and DDR5 migration continue, providing upside.

BEAR CASE NT$300.00

1.1x 2027e BVPS

DDR4 pricing declines drastically: Supply ramp is faster than expected. 1a/1b ramp and DDR5 migration are slower than expected.

Risk Reward – Nanya Technology Corp. (2408.TW)

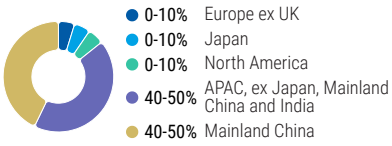
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
12" Wafers Out per quarter (000s)	825.0	820.0	932.5	1,112.5
Average Wafer Yield (%)	81.8	102.8	106.2	95.0
Total Die shipments - 2 gb equivalent (mn)	2,368.0	2,975.7	3,604.0	3,985.1

INVESTMENT DRIVERS

- Specialty DRAM pricing
- Monthly sales momentum
- Industry outlook reported by industry peers

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Sustained DRAM prices from more disciplined supply and stronger demand
- Stronger pricing environment, enabling revenue guidance to be beaten
- Faster-than-expected 1a/1b nm ramp-up progress

RISKS TO DOWNSIDE

- Slower-than-expected 1a/1b nm ramp-up
- Less demand for specialty DRAM from 4K2K TVs and smart set-top boxes

OWNERSHIP POSITIONING

Inst. Owners, % Active 73.3%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	195,412	405,725 / 396,338	427,955
EBITDA (NT$, mn)	131,299	322,470 / 309,696	342,198
Net income (NT$, mn)	97,986	257,292 / 249,970	272,926
EPS (NT$)	72.36	75.96 / 76.43	80.53

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

AP Memory: Estimate revisions summary

We adjust our EPS estimates +21% for 2026, +2% for 2027, and -12% for 2028: We factor in 3Q26 preliminary results. We adjust our 2028 forecasts mainly to account for supply chain risks. We note that the company's IoTRAM is less affected by volatility of typical commodity DRAM pricing. The 2027e and 2028e EPS changes mainly reflect supply chain uncertainties around S-SiCap. We now reduce our S-SiCap unit shipment estimates throughout 2027-2028 and 50%+ into 4Q28. We also slightly reduced our IoT RAM ASP estimates throughout 2027-28 to reflect peaking DDR3/4 pricing power. In addition, we reduced our GM assumptions across product lines to reflect potential peak in pricing power and supply chain uncertainties. Our changes in net income/EPS estimates are largely a function of changes in our revenue and gross profit estimates. We believe it is a good time to make these changes now as a result of the narrower shortages, moderating price hikes, and potential delay in 2028 US cloud capex.

Exhibit 38: AP Memory – Estimate revisions

(NT$ mn)	New '26e	Old '26e	Diff.	New '27e	Old '27e	Diff.	New '28e	Old '28e	Diff.
Net sales	10,980	9,601	14%	20,918	19,356	8%	32,175	34,606	-7%
COGS	5,318	4,883		10,748	9,547		16,203	16,599	
Gross profit	5,661	4,717	20%	10,170	9,809	4%	15,972	18,007	-11%
Operating expenses	1,931	1,749		3,355	3,104		4,851	5,220	
Operating profit	3,730	2,969	26%	6,816	6,705	2%	11,121	12,787	-13%
Non-op. income (exp.)	565	565		437	437		437	437	
Pretax income	4,295	3,534	22%	7,253	7,142	2%	11,558	13,224	-13%
Taxes	878	710		1,451	1,435		2,312	2,656	
Net income	3,485	2,882	21%	5,878	5,763	2%	9,322	10,623	-12%
Reported EPS	**21.39**	**17.69**	**21%**	**36.07**	**35.37**	**2%**	**57.21**	**65.19**	**-12%**
Margins									
Gross margin	51.6%	49.1%	2.4%	48.6%	50.7%	-2.1%	49.6%	52.0%	-2.4%
Operating margin	34.0%	30.9%	3.1%	32.6%	34.6%	-2.1%	34.6%	36.9%	-2.4%
Pretax margin	39.1%	36.8%	2.3%	34.7%	36.9%	-2.2%	35.9%	38.2%	-2.3%
Net margin	31.7%	30.0%	1.7%	28.1%	29.8%	-1.7%	29.0%	30.7%	-1.7%
Opex %	17.6%	18.2%	-0.6%	16.0%	16.0%	0.0%	15.1%	15.1%	0.0%

Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 39: AP Memory – Quarterly financial summary

(NT$ mn)	1Q26	2Q26	3Q26p	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Total Revenues	**2,100**	**2,363**	**3,312**	**3,476**	**4,162**	**4,840**	**5,600**	**6,317**	**7,136**	**7,910**	**8,393**	**8,736**	**5,658**	**10,980**	**20,918**	**32,175**
Sequential Change	*12.4%*	*12.5%*	*40.2%*	*4.9%*	*19.7%*	*16.3%*	*15.7%*	*12.8%*	*13.0%*	*10.8%*	*6.1%*	*4.1%*				
Change vs Year Ago	*115.4%*	*77.8%*	*121.6%*	*86.1%*	*98.2%*	*104.8%*	*69.1%*	*81.7%*	*71.5%*	*63.4%*	*49.9%*	*38.3%*	*35.0%*	*94.1%*	*90.5%*	*53.8%*
Cost of Sales	1,130	1,204	1,457	1,799	2,156	2,497	2,870	3,225	3,624	3,993	4,220	4,365	3,024	5,318	10,748	16,203
Percent of Revenues	*54%*	*51%*	*44%*	*52%*	*52%*	*52%*	*51%*	*51%*	*51%*	*50%*	*50%*	*50%*	*53%*	*48%*	*51%*	*50%*
Gross Profit	**970**	**1,159**	**1,855**	**1,677**	**2,006**	**2,342**	**2,730**	**3,092**	**3,512**	**3,916**	**4,173**	**4,371**	**2,634**	**5,661**	**10,170**	**15,972**
Percent of Revenues	*46.2%*	*49.1%*	*56.0%*	*48.3%*	*48.2%*	*48.4%*	*48.8%*	*48.9%*	*49.2%*	*49.5%*	*49.7%*	*50.0%*	*46.5%*	*51.6%*	*48.6%*	*49.6%*
Incremental Margin	*17%*	*72%*	*73%*	*-108%*	*48%*	*50%*	*51%*	*50%*	*51%*	*52%*	*53%*	*58%*	*33%*	*57%*	*45%*	*52%*
Total Opex	382	434	500	616	712	774	884	984	1,097	1,200	1,256	1,298	1,234	1,931	3,355	4,851
Percent of Revenues	*18.2%*	*18.4%*	*15.1%*	*17.7%*	*17.1%*	*16.0%*	*15.8%*	*15.6%*	*15.4%*	*15.2%*	*15.0%*	*14.9%*	*21.8%*	*17.6%*	*16.0%*	*15.1%*
Operating Income	**588**	**726**	**1,355**	**1,061**	**1,295**	**1,568**	**1,846**	**2,108**	**2,415**	**2,717**	**2,917**	**3,073**	**1,399**	**3,730**	**6,816**	**11,121**
Percent of Revenues	*28.0%*	*30.7%*	*40.9%*	*30.5%*	*31.1%*	*32.4%*	*33.0%*	*33.4%*	*33.8%*	*34.3%*	*34.8%*	*35.2%*	*24.7%*	*34.0%*	*32.6%*	*34.6%*
Total Non-operating Income (Loss)	237	109	109	109	109	109	109	109	109	109	109	109	119	565	437	437
Profit Before Taxes	826	835	1,467	1,170	1,404	1,677	1,955	2,217	2,524	2,826	3,026	3,182	1,518	4,295	7,253	11,558
Percent of Revenues	*39%*	*35%*	*44%*	*34%*	*34%*	*35%*	*35%*	*35%*	*35%*	*36%*	*36%*	*36%*	*27%*	*39%*	*35%*	*36%*
Taxes	166	168	310	234	281	335	391	443	505	565	605	636	279	878	1,451	2,312
Tax Rate	*20.1%*	*20.1%*	*21.2%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*18.3%*	*20.4%*	*20.0%*	*20.0%*
Reported Income (TW GAAP)	**676**	**681**	**1,176**	**955**	**1,142**	**1,361**	**1,583**	**1,792**	**2,038**	**2,280**	**2,440**	**2,564**	**1,258**	**3,485**	**5,878**	**9,322**
Percent of Revenues	*32.2%*	*28.8%*	*35.5%*	*27.5%*	*27.4%*	*28.1%*	*28.3%*	*28.4%*	*28.6%*	*28.8%*	*29.1%*	*29.4%*	*22.2%*	*31.7%*	*28.1%*	*29.0%*
Change vs Year Ago	*104%*	*-225%*	*66%*	*25%*	*69%*	*100%*	*35%*	*88%*	*78%*	*68%*	*54%*	*43%*	*-20%*	*177%*	*69%*	*59%*
Reported EPS (NT$, TW GAAP)	**4.15**	**4.18**	**7.21**	**5.86**	**7.01**	**8.35**	**9.71**	**11.00**	**12.51**	**13.99**	**14.97**	**15.74**	**7.74**	**21.39**	**36.07**	**57.21**
Change vs Year Ago	*103%*	*-224%*	*66%*	*24%*	*69%*	*100%*	*35%*	*88%*	*78%*	*68%*	*54%*	*43%*	*-20%*	*177%*	*69%*	*59%*

Source: Company data, Morgan Stanley Research (e) estimates

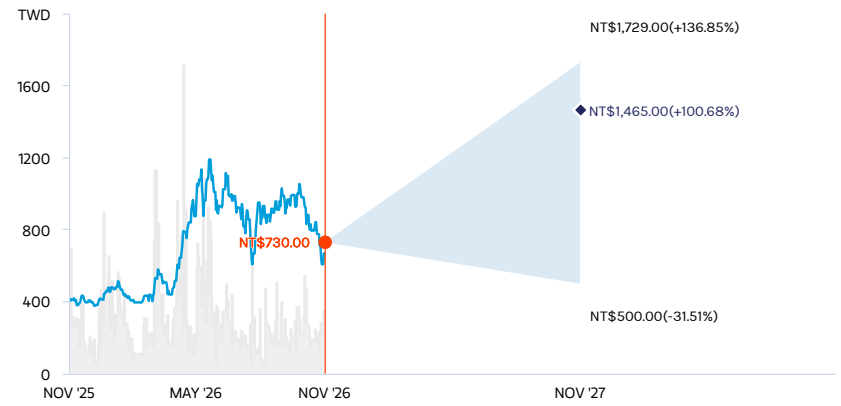
Risk Reward – AP Memory Technology Corp (6531.TW)

Multiple growth drivers ahead; OW

PRICE TARGET NT$1,465.00

Base case, derived from a residual income model. Key RI model assumptions include cost of equity constant at 9.2% (2.0% risk-free rate, 6.0% risk premium, 1.2 beta); medium-term growth rate of 16.0%; terminal growth rate of 3.0%; and payout ratio of 67%.

RISK REWARD CHART



Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- S-SiCap revenue is likely to ramp in 2026-27, and be further driven by new IPD interposer opportunities into 2028.
- Memory bandwidth and power consumption are the next issues to be resolved in AI computing, and we believe AP Memory's wafer-on wafer (WoW) packaging technology could be an effective solution.
- The company expects memory business to be stable in the near term.
- Overall AI business should continue to rise in the mix – a positive read for the margin profile.
- Our price target implies 41x 2027e P/E, roughly in line with the historical average since 2020. On this basis, we find valuation attractive.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*
Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$1,729.00

48x 2027e EPS

Faster-than-expected AI and IPD business progress with PSRAM generating more cash flow: We assume: 1) WoW packaging revenue to reach >NT$10bn, and S-SiCap revenue to reach >NT$10bn by 2028; 2) total revenue to rise at a 55%+ CAGR, 2025-28; and 3) gross margin to increase from 46.5% in 2025 to 60%+ by 2028.

BASE CASE NT$1,465.00

41x 2027e EPS

Fast-growing AI and IPD business, with PSRAM generating stable cash flows: We expect: 1) WoW revenue to reach NT$11bn+, and S-SiCap revenue to reach NT$6bn+ by 2028; 2) total revenue to rise at a ~80% CAGR, 2025-28; and 3) gross margin to increase from 46.5% in 2025 to ~50% in 2028.

BEAR CASE NT$500.00

14x 2027e EPS

Slower-than-expected AI and IPD business progress, with PSRAM facing intensifying competition: We assume: 1) WoW packaging revenue of ~NT$1.5bn, and S-SiCap revenue of ~NT$2bn by 2028; 2) total revenue to rise at a 10-15% CAGR, 2025-28; and 3) gross margin to remain at ~51% by 2028.

Risk Reward – AP Memory Technology Corp (6531.TW)

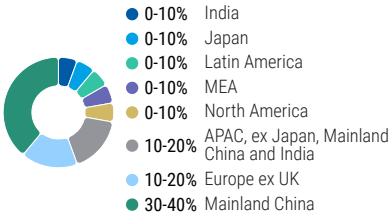
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
1. Wifi Sales (%) (%)	73.7	58.4	55.0	47.5
2. Ethernet Sales (%) (%)	7.1	8.1	19.8	32.8

INVESTMENT DRIVERS

- Demand for AI applications
- Consumer tech demand, especially from the IoT market

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger-than-expected consumer demand
- Faster-than-expected IPD ramp and WoW packaging development
- Milder competition from partners (foundry, memory house, GPU vendor) or other design houses

RISKS TO DOWNSIDE

- Weaker-than-expected consumer demand
- Slower-than-expected IPD ramp and WoW packaging development
- Intensifying competition from partners (foundry, memory house, GPU vendor) or other design houses

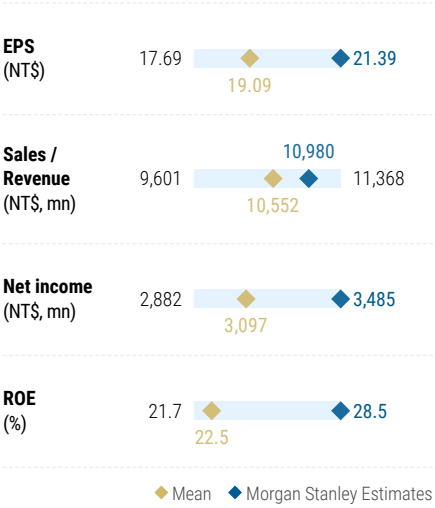
OWNERSHIP POSITIONING

Inst. Owners, % Active 64.4%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

Phison: Estimate Revisions

We adjust our 2026, 2027, and 2028 EPS estimates: We factor in 3Q26 results, which demonstrated continued revenue growth (+27% Q/Q) but also gross margin decline (59.7% in 3Q, down from 65.3% in 2Q). In 4Q26 and 2027, we expect emerging weakness from consumer tech and a more moderate pricing environment to result in 4Q revenue decline, with GM starting to normalize. We think normalized GM is likely to extend into 2028.

We present further discussion of our EPS estimates under Valuation Methodology.

Exhibit 44: Phison: Quarterly financials

NT$ in million	1Q26	2Q26	3Q26P	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Total Revenues	**40,967**	**67,888**	**86,076**	**67,390**	**54,144**	**55,957**	**53,692**	**46,198**	***58,936***	***72,664***	***262,321***	***209,990***	***155,219***
Sequential Change	*79.7%*	*65.7%*	*26.8%*	*-21.7%*	*-19.7%*	*3.3%*	*-4.0%*	*-14.0%*					
Change vs Year Ago	*196.0%*	*279.5%*	*374.6%*	*195.6%*	*32.2%*	*-17.6%*	*-37.6%*	*-31.4%*	*22.2%*	*23.3%*	*261.0%*	*-19.9%*	*-26.1%*
Cost of Sales	15,849	23,554	34,689	27,666	23,839	26,228	26,695	24,272	39,837	47,806	101,758	101,034	89,968
Percent of Revenues	*39%*	*35%*	*40%*	*41%*	*44%*	*47%*	*50%*	*53%*	*68%*	*66%*	*39%*	*48%*	*58%*
Gross Profit	**25,118**	**44,335**	**51,387**	**39,724**	**30,305**	**29,729**	**26,997**	**21,926**	***19,099***	***24,858***	***160,563***	***108,956***	***65,251***
Percent of Revenues	*61.3%*	*65.3%*	*59.7%*	*58.9%*	*56.0%*	*53.1%*	*50.3%*	*47.5%*	*32.4%*	*34.2%*	*61.2%*	*51.9%*	*42.0%*
Incremental Margin	*86%*	*71%*	*39%*	*NM*	*NM*	*-32%*	*NM*	*NM*	*28%*	*42%*	*72%*	*NM*	*NM*
Total Opex	10,277	17,964	18,489	15,014	12,864	13,854	13,830	12,362	*15,566*	*16,591*	*61,744*	*52,909*	*45,285*
Percent of Revenues	*25.1%*	*26.5%*	*21.5%*	*22.3%*	*23.8%*	*24.8%*	*25.8%*	*26.8%*	*26.4%*	*22.8%*	*23.5%*	*25.2%*	*29.2%*
R&D	8,671	15,099	14,855	12,304	10,427	11,336	11,414	10,283	*12,586*	*13,775*	*50,929*	*43,459*	*38,300*
Percent of Revenues	*21.2%*	*22.2%*	*17.3%*	*18.3%*	*19.3%*	*20.3%*	*21.3%*	*22.3%*	*21.4%*	*19.0%*	*19.4%*	*20.7%*	*24.7%*
General & administrative	830	1,577	2,000	1,566	1,245	1,287	1,235	1,063	*1,286*	*1,169*	*5,972*	*4,830*	*3,570*
Percent of Revenues	*2.0%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.2%*	*1.6%*	*2.3%*	*2.3%*	*2.3%*
Selling & marketing	776	1,289	1,634	1,144	1,191	1,231	1,181	1,016	*1,694*	*1,647*	*4,843*	*4,620*	*3,415*
Percent of Revenues	*1.9%*	*1.9%*	*1.9%*	*1.7%*	*2.2%*	*2.2%*	*2.2%*	*2.2%*	*2.9%*	*2.3%*	*1.8%*	*2.2%*	*2.2%*
Operating Income	**14,841**	**26,370**	**32,898**	**24,710**	**17,442**	**15,875**	**13,167**	**9,564**	**3,533**	**8,266**	**98,819**	**56,047**	**19,967**
Percent of Revenues	*36.2%*	*38.8%*	*38.2%*	*36.7%*	*32.2%*	*28.4%*	*24.5%*	*20.7%*	*6%*	*11%*	*38%*	*27%*	*13%*
Change vs Year Ago	*N/A*	*N/A*	*2209.8%*	*641.0%*	*17.5%*	*-39.8%*	*-60.0%*	*-61.3%*	*-2%*	*134%*	*1095%*	*-43%*	*-64%*
Total Non-operating Income(Loss)	3,057	4,704	3,077	183	183	183	183	183	6,216	1,939	11,022	732	732
Profit Before Taxes	**17,898**	**31,075**	**35,976**	**24,893**	**17,624**	**16,058**	**13,350**	**9,747**	**9,749**	**10,206**	**109,841**	**56,779**	**20,698**
Percent of Revenues	*44%*	*46%*	*42%*	*37%*	*33%*	*29%*	*25%*	*21%*	*17%*	*14%*	*42%*	*27%*	*13%*
Taxes	2,723	4,856	5,428	4,481	3,172	2,890	2,403	1,754	*1,795*	*1,465*	*17,487*	*10,220*	*3,726*
Tax Rate	*15.2%*	*15.6%*	*15.1%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18%*	*14%*	*16%*	*18%*	*18%*
Net Income, Cont Ops	15,175	26,219	30,548	20,412	14,452	13,168	10,947	7,993	7,954	8,741	92,354	46,559	16,973
Percent of Revenues	*37%*	*39%*	*35%*	*30%*	*27%*	*24%*	*20%*	*17%*	*13%*	*12%*	*35%*	*22%*	*11%*
Net Income (incl. Emp. Bonus Expense)	15,175	26,219	30,548	20,412	14,452	13,168	10,947	7,993	7,955	8,743	92,357	46,559	16,973
Percent of Revenues	*37%*	*39%*	*35%*	*30%*	*27%*	*24%*	*20%*	*17%*	*13.5%*	*12.0%*	*35.2%*	*22.2%*	*10.9%*
	0%	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*
Reported Income (TW GAAP)	**15,175**	**26,219**	**30,548**	**20,412**	**14,452**	**13,168**	**10,947**	**7,993**	**7,955**	**8,743**	**92,357**	**46,559**	**16,973**
Percent of Revenues	*37%*	*39%*	*35%*	*30%*	*27%*	*24%*	*20%*	*17%*	*13%*	*12%*	*35%*	*22%*	*11%*
Change vs Year Ago	*0%*	*73%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*119%*	*10%*	*956%*	*-50%*	*-64%*
Reported EPS (NT$, TW GAAP)	**68.86**	**118.97**	**138.62**	**92.62**	**65.58**	**59.75**	**49.67**	**36.27**	**35.68**	**39.70**	**419.09**	**211.27**	**77.02**
Change vs Year Ago	*1233%*	*3406%*	*1272%*	*341%*	*-5%*	*-50%*	*-64%*	*-61%*	*102%*	*11%*	*956%*	*-50%*	*-64%*

Source: Company data, Morgan Stanley Research (E) estimates

Phison: Valuation Methodology

We cut our price target from NT$2,588 to NT$1,357

Our price target is our base case value, derived from a residual income (RI) model. Our cut reflects our latest earnings estimates. We also cut our terminal growth rate assumption from 4.0% to 3.0% to be broadly in line with a conservative estimate of long-term nominal economic growth in developed economies. We believe this is appropriate for Phison because rising storage demand should support continued growth, while declining NAND prices per bit, competition, and maturing SSD penetration should limit the conversion of volume growth into earnings growth, bringing its growth closer to that of the broader economy over time.

Our other RI parameters are unchanged, including a cost of equity of 9.0%, a medium-term growth rate of 9.0%, and a dividend payout ratio assumption of 50.0%.

Apart from the earnings estimate revisions and terminal growth rate change, a material part of the price target change is driven by an error in the previously published residual income model (August 2026). The error has now been corrected. In the previous RI model, the 2028e net income was mistakenly calculated as 2027e net income * (1+ intermediate growth rate), where it should have been linked to the actual 2028e net income forecast. This overstated the previous price target.

Our earnings estimates have also changed materially from the previously published version: they increase 12% for 2026, 60% for 2027, and 37% for 2028. This mainly reflects higher GM assumptions of 61.2%, 51.9%, and 42.0%, respectively; previously we assumed 60.1%, 43.4%, and 39.5%.

Our base case value/price target is lowered 48%, of which roughly +37% is contributed by earnings estimate revisions, -6% is contributed by terminal growth rate reduction, and -83% is contributed by the correction of the prior RI model error. (The overall price target change is calculated from the fully revised RI model. The individual contribution estimates are illustrative and should not be added together or compounded to derive the overall change.)

We cut our bull and bear case values from NT$3,400 and NT$1,250 to NT$1,782 and NT $655, respectively, implying 8.4x 2027e EPS in the bull case and 3.1x 2027e EPS in the bear case.

We downgrade Phison to UW

The company hopes its AiDaptive module and service business can pick up and offset the margin pressure in the conventional module business in the coming quarters, which we believe is worth monitoring. However, its 3Q GM of 59% missed the Street estimate of 60.3%; we see this as a potential sign that the previously strong margin profile supported by low-cost inventory may start to face more downside. Specifically, the AI storage part of its business (eSSD, and cSSD for AI PC, together 40% of revenue) may face margin decline. This is what we have been cautious about, and we see the 3Q margin miss as having provided a good opportunity to downgrade this stock, which we view as

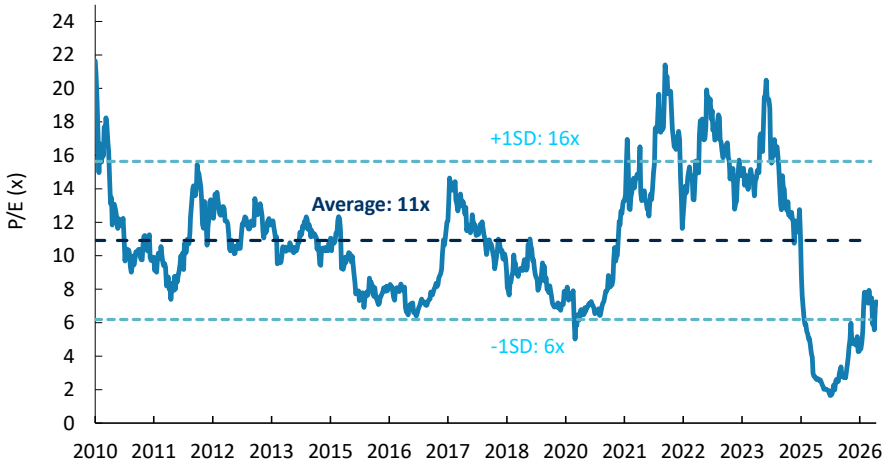
overvalued based on our revised assumptions.

Exhibit 45: Phison: RI model

NT$million	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2035E
Total Equity	**147,997**	**147,388**	**140,583**	**150,758**	**161,849**	**173,938**	**187,115**	**201,478**	**217,134**	**234,198**	**252,799**
Net Profit	**92,357**	**46,559**	**16,973**	**18,500**	**20,165**	**21,980**	**23,958**	**26,115**	**28,465**	**31,027**	**33,819**
ROAE	**88.8%**	**31.5%**	**11.8%**	**12.7%**	**12.9%**	**13.1%**	**13.3%**	**13.4%**	**13.6%**	**13.7%**	**13.9%**
Residual Income	47,988	33,448	4,221	5,308	5,996	6,745	7,562	8,451	9,420	10,477	11,627
Spread	**79.8%**	**22.6%**	**2.9%**	**3.8%**	**4.0%**	**4.2%**	**4.3%**	**4.5%**	**4.7%**	**4.8%**	**5.0%**
Ending Equity Capital	147,997										
PV of Forecast Period	65,813										
PV of Continuing Value	85,995										
Equity Value	**299,805**										
No. of Shares	221										
Projected Price	**1,357**										

Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 46: Phison: P/E multiple



Source: Company data, Morgan Stanley Research estimates

Risk Reward – Phison Electronics Corp (8299.TWO)

AI still drives NAND market, but caution on GM decline; UW

PRICE TARGET NT$1,357.00

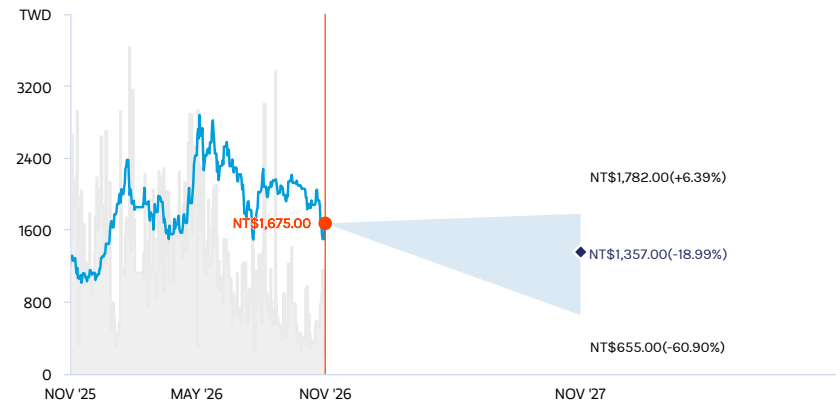
Base case, residual income model. Key assumptions: a cost of equity of 9.0%, a medium-term growth rate of 9.0%, a terminal growth rate of 3.0%, and a dividend payout ratio assumption of 50%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



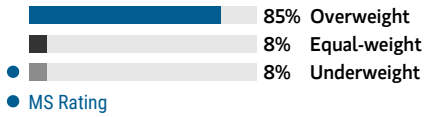
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

- AI-driven storage demand is extending the industry up-cycle, but we see limited upside for module vendors given constrained NAND supply, FIFO-driven margin normalization, and weakening consumer SSD demand amid elevated NAND prices.
- eSSD and AI-related storage form meaningful growth opportunities for Asian module house vendors such as Phison. Even so, we believe the scale remains insufficient to offset broader consumer SSD weakness.
- We see risks in execution as raw NAND suppliers are looking to cut support further to module houses to prioritize their other customers. Industry average for shipments is -40% YoY, which could create risks for Phison.
- We find valuation unattractive.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

BULL CASE NT$1,782.00

8.4x 2027e EPS

Growing SSD demand and limited supply growth prompt NAND flash shortages; SSD market growth accelerates: 1) Revenue increases at a CAGR of >40% over 2025-28; 2) gross margin is at 50-60% in 2028 vs. 61% in 2026.

BASE CASE NT$1,357.00

6.4x 2027e EPS

A good year for NAND modules in 2026, but we doubt it can be sustained into 2027: 1) Revenue increases at a CAGR of 29% over 2025-28e; 2) gross margin decreases to 42% in 2028 from 61% in 2026, as NAND pricing normalizes and low cost inventory is depleted.

BEAR CASE NT$655.00

3.1x 2027e EPS

No outsourcing from Micron, low SSD market growth, NAND flash prices drop more than in our base case: 1) Revenue increases at a CAGR of <10% in 2025-28; 2) gross margin faces more significant erosion, to <30% in 2028 from 61% in 2026, as NAND pricing decreases faster than expected.

Risk Reward – Phison Electronics Corp (8299.TWO)

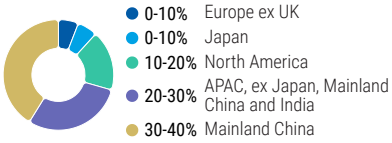
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Flash module revenue Y/Y (%)	22.0	354.0	(19.0)	(28.1)
Flash controller IC revenue Y/Y (%)	44.7	22.4	(28.2)	(3.1)
Flash module gross margin (%)	33.5	63.0	52.3	41.8
Flash controller IC gross margin (%)	43.9	44.2	43.2	43.3

INVESTMENT DRIVERS

- Rising NAND flash prices
- Higher SSD penetration rate
- Adoption of solution by key OEMs

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected eMMC controller IC outsourcing rate
- Faster SSD market growth
- Market share gain in ultrabooks and SSDs

RISKS TO DOWNSIDE

- Weaker-than expected NAND demand
- Milder impact on global NAND supply
- Non-acceptance of price hikes on NAND modules
- Risks in execution as raw NAND suppliers reprioritize

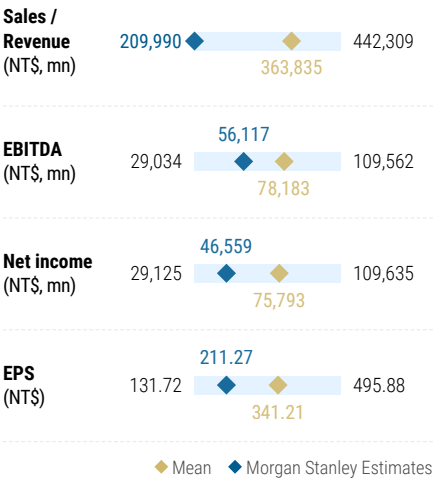
OWNERSHIP POSITIONING

Inst. Owners, % Active 50.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



Source: Refinitiv, Morgan Stanley Research

Phison: Financial Summary

Exhibit 47: Financial summary

Income Statement

NT$mn (Years End Dec)	2024	2025E	2026E	2027E	2028E
Net sales	58,936	72,664	262,321	209,990	155,219
COGS	(39,837)	(47,806)	(101,758)	(101,034)	(89,968)
Gross profit	19,099	24,858	160,563	108,956	65,251
Operating expenses	(15,566)	(16,591)	(61,744)	(52,909)	(45,285)
Operating income	**3,533**	**8,266**	**98,819**	**56,047**	**19,967**
Non-operating income	6,216	1,939	11,022	732	732
Pre-tax income	9,749	10,206	109,841	56,779	20,698
Income tax	1,795	1,465	17,487	10,220	3,726
Reported net Income	**7,954**	**8,741**	**92,354**	**46,559**	**16,973**
Adj.wtd.avg.shrs(m)	204	207	221	221	221
Reported EPS (NT$)	**35.68**	**39.70**	**419.09**	**211.27**	**77.02**

Balance Sheet

NT$mn (Years End Dec)	2024	2025E	2026E	2027E	2028E
Cash	19,982	16,275	95,510	101,461	103,163
Mkt Securities	784	1,475	1,475	1,475	1,475
AR/NR	8,011	13,689	32,804	26,259	19,410
Inventory	24,614	35,609	28,985	28,778	25,626
Other	649	1,432	1,432	1,432	1,432
Current Assets	**54,041**	**68,480**	**160,205**	**159,405**	**151,106**
Long-term investments	651	1,171	1,171	1,171	1,171
Fixed assets	3,783	3,883	3,983	4,083	4,183
Deferred assets	701	629	629	629	629
Other assets	10,164	11,593	11,593	11,593	11,593
Total Assets	**69,339**	**85,756**	**177,581**	**176,881**	**168,682**
S/T borrowings	41	0	0	0	0
AP/NP	1,309	8,878	12,814	12,723	11,329
Other ST liabilities	12,870	16,417	16,417	16,417	16,417
LT debt	6,054	352	352	352	352
Other LT liabilities	0	0	0	0	0
Common shares	2,058	2,180	2,180	2,180	2,180
Total Liabilities	**20,273**	**25,648**	**29,584**	**29,493**	**28,099**
Additional capital	13,054	19,104	19,104	19,104	19,104
Retained earning	34,283	39,151	127,040	126,432	119,626
Other shareholders' equity	(329)	(327)	(327)	(327)	(327)
Total Equity	**49,066**	**60,108**	**147,997**	**147,388**	**140,583**
Total Liab. & Shrhldr's Equity	**69,339**	**85,756**	**177,581**	**176,881**	**168,682**

E = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, Company Data

Cash Flow Statement

NT$mn (Years End Dec)	2024	2025E	2026E	2027E	2028E
Cashflow from Operations	**2,094**	**1,598**	**83,872**	**53,288**	**25,650**
Net profits	7,955	8,743	92,357	46,559	16,973
Depreciation	695	755	70	70	70
Working Capital Change	(3,895)	(9,886)	(8,555)	6,659	8,608
Other adjustments	(2,661)	1,986	0	0	0
Cashflow from Investing	**3,903**	**(1,908)**	**(170)**	**(170)**	**(170)**
Capex	(955)	(1,397)	(170)	(170)	(170)
Change of LT Investment	(705)	(529)	0	0	0
Change of ST Investment	(53)	(402)	0	0	0
Other adjustments	5,616	420	0	0	0
Cashflow from financing	**(673)**	**(3,109)**	**(4,468)**	**(47,167)**	**(23,778)**
Increase in L/T debt	5,681	(5,701)	0	0	0
Increase in S/T debt	41	(41)	0	0	0
Cash Dividend Paid	(2,661)	(5,172)	(4,468)	(47,167)	(23,778)
Dir& Emp Bonus Paid	0	0	0	0	0
Issuance of stock	0	0	0	0	0
Other adjustments	(3,733)	7,806	0	0	0
Exchange rate adjustment	437	(288)	0	0	0
Net change in cash	**5,762**	**-3,707**	**79,235**	**5,951**	**1,702**

Financial Ratios

	2024	2025E	2026E	2027E	2028E
Growth(%)					
Turnover	22.2	23.3	261.0	-19.9	-26.1
Operating profits	-2.5	134.0	1095.4	-43.3	-64.4
Pretax profits	137.8	4.7	976.3	-48.3	-63.5
Net profits	119.5	9.9	956.5	-49.6	-63.5
EPS	102.4	11.3	955.7	-49.6	-63.5
Margins (%)					
Gross Margin	32.4	34.2	61.2	51.9	42.0
Operating Margin	6.0	11.4	37.7	26.7	12.9
Pretax Margin	16.5	14.0	41.9	27.0	13.3
Net Profit	13.5	12.0	35.2	22.2	10.9
Return (%)					
ROAE	17.3	16.0	88.8	31.5	11.8
ROAA	11.8	11.3	70.1	26.3	9.8
Gearing (%)					
Net Debt/Equity	(28.3)	(26.5)	(64.3)	(68.6)	(73.1)
Liabilities/Equity	41.3	42.7	20.0	20.0	20.0
Ratios (X)					
Current ratio	3.8	2.7	5.5	5.5	5.4
Quick ratio	2.0	1.2	4.4	4.4	4.4
Others					
AR/NR Turnover (days)	46	46	46	46	46
Inventory Turnover (days)	104	104	104	104	104
AP Turnover (days)	46	46	46	46	46
Cash Conversion (days)	104	104	104	104	104

Source: Morgan Stanley Research (forecasts), Company Data

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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Global Stock Ratings Distribution

(as of October 31, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1575	43%	460	49%	29%	742	44%
Equal-weight/Hold	1537	42%	393	42%	26%	720	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	92	10%	16%	218	13%
Total	3,684		946			1681	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (11/10/2026)
Charlie Chan		
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT$99.90
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT$3,420.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT$605.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb838.00

CXMT Corp (688825.SS)	O (08/26/2026)	Rmb40.45
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT$7,170.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT$421.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK$74.55
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK$70.10
MediaTek (2454.TW)	O (11/28/2025)	NT$3,830.00
MetaX Integrated Circuits (688802.SS)	U (11/02/2026)	Rmb509.03
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT$383.00
Phison Electronics Corp (8299.TWO)	U (11/10/2026)	NT$1,675.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb81.55
Silergy Corp. (6415.TW)	U (05/19/2026)	NT$440.50
SMIC (0981.HK)	O (10/21/2025)	HK$47.24
TSMC (2330.TW)	O (02/07/2022)	NT$2,285.00
UMC (2303.TW)	O (05/19/2026)	NT$100.50
Vanguard International Semiconductor (5347.TWO)	U (10/30/2026)	NT$132.50
WIN Semiconductors Corp (3105.TWO)	O (10/29/2026)	NT$484.50
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK$96.20
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb49.08
Empyrean Technology Co Ltd (301269.SZ)	O (11/02/2026)	Rmb90.47
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb23.97
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb190.00
Innoscience (2577.HK)	E (10/13/2025)	HK$27.36
JCET Group Co Ltd (600584.SS)	O (09/16/2026)	Rmb52.78
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK$23.50
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb79.95
SJ Semiconductor Corp (688820.SS)	E (09/16/2026)	Rmb81.82
StarPower Semiconductor Ltd (603290.SS)	U (10/31/2026)	Rmb70.81
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb51.85
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb19.37
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb75.75
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT$730.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT$1,125.00
Aspeed Technology (5274.TWO)	E (11/10/2026)	NT$11,935.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT$84.10
Elan Microelectronics Corp (2458.TW)	E (11/04/2026)	NT$117.00
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT$2,215.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb176.15
GigaDevice Semiconductor Beijing Inc (603986.SS)	E (11/10/2026)	Rmb244.57
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US$8.61
Macronix International Co Ltd (2337.TW)	E (11/10/2026)	NT$82.10
Montage Technology Co Ltd (6809.HK)	E (11/10/2026)	HK$190.80
Montage Technology Co Ltd (688008.SS)	E (11/10/2026)	Rmb154.80
Novatek (3034.TW)	U (02/04/2026)	NT$477.50
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT$103.50
Parade Technologies Ltd (4966.TWO)	U (10/29/2026)	NT$397.50
Powerchip Semiconductor Manufacturing Co (6770.TW)	E (11/10/2026)	NT$59.20
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT$745.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb63.23
Silicon Motion (SIMO.O)	O (05/06/2024)	US$172.01
Winbond Electronics Corp (2344.TW)	E (11/10/2026)	NT$135.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT$92.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT$190.00
Henry Zhao		

ACM Research Inc (ACMR.O)	O (03/07/2023)	US$49.75
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb245.66
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb55.43
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb538.68
OmniVision Integrated Circuits Group Inc (603501.SS)	E (10/29/2026)	Rmb87.88
Lucas Wang		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT$1,025.00
FOCI Fiber Optic Communications Inc (3363.TWO)	E (10/09/2026)	NT$411.00
GlobalWafers Co Ltd (6488.TWO)	E (11/04/2026)	NT$612.00
Hon Precision (7769.TW)	O (04/17/2026)	NT$4,220.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT$227.50
MPI Corporation (6223.TWO)	O (04/17/2026)	NT$4,075.00

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